

SUMS OF TWO SQUARES IN THREE-TERM PATTERNS, II: AUTOMORPHIC KERNELS AND AN IMPROVED EXPONENT

ABSTRACT. For fixed integers $0 < a < b$ let $S_{a,b}(x)$ be the number of $n \leq x$ such that $n, n+a$ and $n+b$ are all sums of two squares. In an earlier paper we showed that $S_{a,b}(x) \gg x^{1/2+\delta}$ for every $\delta < 1/24$. Here we prove this for every $\delta < (1-2\theta)/(10-12\theta)$, where θ is any exponent towards Selberg's eigenvalue conjecture for congruence subgroups. The Kim–Sarnak bound $\theta = 7/64$ gives every $\delta < 25/278 = 0.0899\dots$, and Selberg's conjecture would give every $\delta < 1/10$.

As before, two of the three numbers are made sums of two squares by an identity, and the third runs through the values of a quadratic polynomial attached to a prime d . We change the families so that every 2-adic condition falls on d . The third number then becomes $2^{v-2}(E\ell^2 + H_d)/d$, where ℓ runs over all integers, $E \in \{4, 8\}$ and $H_d \asymp d^2$. Its r -weighted count leads to Type I sums for the roots of $E\ell^2 + H_d \equiv 0$ modulo dk with $k \equiv \pm 1 \pmod{4}$. We parametrise these roots by Heegner points of discriminant $-4EH_d$ on $\Gamma_0(4Ed)$ and estimate the sums with a bound of Grimmelt and Merikoski for weighted averages of automorphic kernels, together with bounds for the two resulting kernels at a single level. This replaces Weil's bound for Kloosterman sums, which was used in the earlier work, and admits primes d up to $x^{(1-2\theta)/(5-6\theta)-\varepsilon}$.

1. INTRODUCTION

Let $\mathcal{S} = \{x^2 + y^2 : x, y \in \mathbb{Z}\}$. For fixed integers $0 < a < b$ put

$$S_{a,b}(x) = \#\{1 \leq n \leq x : n, n+a, n+b \in \mathcal{S}\}.$$

Hooley [10] proved that $S_{a,b}(x) \rightarrow \infty$, answering a question of Littlewood. Freiberg, Kurlberg and Rosenzweig [6, Conjecture 1.1] formulated an analogue of the Hardy–Littlewood conjecture for sums of two squares. It predicts $S_{a,b}(x) \sim \mathfrak{S}_{\{0,a,b\}} C^3 x (\log x)^{-3/2}$, where C is the Landau–Ramanujan constant and $\mathfrak{S}_{\{0,a,b\}} > 0$ is a singular series. Upper bounds of this order follow from sieve methods [1, 15], but lower bounds are much weaker. We refer to [11, §1] for a discussion, and to [14] for the triple correlation of the divisor function, where by contrast the order of magnitude is known. In [11] we proved that $S_{a,b}(x) \gg x^{1/2+\delta}$ for every $\delta < 1/24$; no bound with $\delta > 0$ had previously been known for any pattern. The purpose of this paper is to improve the exponent.

For a congruence subgroup $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$, let $\lambda_1(\Gamma)$ be the smallest positive eigenvalue of the Laplacian on $\Gamma \backslash \mathbb{H}$, and put $\theta(\Gamma) = \max\{0, \mathrm{Re} \sqrt{1/4 - \lambda_1(\Gamma)}\}$. We call $\theta \in [0, 1/2)$ *admissible* if $\theta(\Gamma_0(q)) \leq \theta$ for every $q \geq 1$. Selberg's eigenvalue conjecture states that $\theta = 0$ is admissible, and Kim and Sarnak [13, Appendix 2] proved that $\theta = 7/64$ is admissible.

Theorem 1.1. *Let $0 < a < b$ be fixed integers and let θ be admissible. For every*

$$\delta < \delta_\theta := \frac{1-2\theta}{10-12\theta}$$

we have $S_{a,b}(x) \gg_{a,b,\delta} x^{1/2+\delta}$ for $x \geq x_0(a, b, \delta)$.

Corollary 1.2. *For every $\delta < 25/278 = 0.0899\dots$ we have $S_{a,b}(x) \gg_{a,b,\delta} x^{1/2+\delta}$. If Selberg's eigenvalue conjecture holds, then this is true for every $\delta < 1/10$. In particular, for every $\delta < 25/278$ there are $\gg_\delta x^{1/2+\delta}$ integers $n \leq x$ such that $n-1, n$ and $n+1$ are all sums of two squares.*

The corollary is the case $\theta = 7/64$, respectively $\theta = 0$, of Theorem 1.1. For comparison, $1/24 = 0.0417\dots$

The proof relies on a bound of Grimmelt and Merikoski for weighted averages of automorphic kernels, Theorem 8.1 of their preprint [8], in the form stated in [9, Theorem 2.1]; see Theorem 6.1 below. The implied constants in Theorem 1.1 are ineffective because Siegel's theorem is used; Remark 8.4 explains how to avoid this.

1.1. The method. We recall the approach of [11]. Two of the three numbers are made sums of two squares automatically. Let B be odd and put $Q = 2P + B$. If

$$2dQ = u^2 + d^2 + B^2$$

for integers d, u , then $4P(P + B) = (Q - d)^2 + u^2$. If d is moreover a prime with $d \equiv 1 \pmod{4}$, then P and $P + B$ are both sums of two squares (Lemma 2.5). The third number $T = P + A$ then satisfies

$$4dT = u^2 + m_d, \quad m_d = d^2 + 2cd + B^2, \quad c = 2A - B.$$

For a fixed prime d and u in a lattice, T runs through the values of a quadratic polynomial, and $\asymp \sqrt{x/d}$ of these are at most x . We show that the r -weighted count of these values has the expected size. Summing over primes $d \leq x^\varpi$ and removing multiplicities at a cost of x^ε then gives $S_{a,b}(x) \gg x^{1/2+\varpi/2-\varepsilon}$.

The weighted count is evaluated by the hyperbola method. This leads to Type I sums

$$\sum_{k \leq \sqrt{x}} \chi(k) \#\{u : dk \mid u^2 + m_d, \dots\}$$

for the roots of a quadratic congruence. The moduli dk go up to about $d\sqrt{x}$, which exceeds the length $\asymp \sqrt{dx}$ of the u -sum by a factor $\asymp \sqrt{d}$. In [11] these sums were treated by Hooley's method, via Gauss's correspondence and Weil's bound for Kloosterman sums. The resulting error term $x^{3/8+\varepsilon}d$ restricts d to $d \leq x^{1/12-\varepsilon}$ and gives $\delta < 1/24$.

Here we use the spectral theory of automorphic forms instead, in the form developed by Grimmelt and Merikoski [8, 9]. Following Duke, Friedlander and Iwaniec [4], the roots of $E\ell^2 + H \equiv 0 \pmod{\mu}$, for all moduli μ at once, correspond to the integral binary quadratic forms of determinant $N = EH$, that is, to Heegner points of discriminant $-4N$. If the moduli are restricted by a congruence condition that is invariant under $\Gamma_0(q)$, the Type I sum becomes the pairing of an automorphic kernel for $\Gamma_0(q)$ with a weighted sum over Heegner points. Theorem 2.1 of [9] bounds such pairings by two simpler kernels. In [9] these kernels are estimated on average over the level q , by a divisor-switching argument that needs the polynomial to stay fixed while the level varies. In our families the polynomial changes with d , and each d comes with a single level $q = 4Ed$. We therefore estimate the kernels at a single level (Lemmas 6.3 and 6.4). The outcome, Proposition 6.5, is a Type I estimate with square-root cancellation in the length X of the sum. The price is a factor $H^{1/4} \asymp d^{1/2}$, together with a factor $(1 + X/(dH^{1/2}))^\theta$ from possible exceptional eigenvalues.

To apply this, the sums must have exactly this shape. The variable ℓ must run over all integers with a smooth weight, and the conditions on the moduli must be $\Gamma_0(q)$ -invariant. In [11] the 2-adic conditions that make T a norm from $\mathbb{Q}_2(i)$ were imposed through a congruence $u \equiv u_0 \pmod{2^j}$, with $u_0 \neq 0$ in general, and this destroys the structure. We therefore change the families. The automatic pair is chosen with odd difference B (Lemma 2.3), and all 2-adic conditions are imposed on d alone. By Proposition 3.2 there are $v \geq 2$ and d_0 with the following property: for every prime $d \equiv d_0 \pmod{2^{v+2}}$ and every $u \equiv 0 \pmod{2^j}$ with $d \mid u^2 + B^2$, where $j = \lceil (v+2)/2 \rceil$, the number T is 2^{v-2} times an integer $\equiv 1 \pmod{4}$. Writing $u = 2^j \ell$, the polynomial becomes $2^v(E\ell^2 + H_d)$ with $E \in \{4, 8\}$ and $H_d = 2^{-v}m_d$. The moduli dk with $k \equiv \kappa \pmod{4}$ correspond to the $\Gamma_0(4Ed)$ -invariant condition $\mathfrak{c} \equiv \kappa Ed \pmod{4Ed}$ on symmetric matrices $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$. Finding

such data for every pattern requires the 2-adic analysis of §3, and it dictates the choice of the pair in Lemma 2.3.

In the resulting asymptotic formula for the weighted count, the main term is $\gg x^{1/2}d^{-1/2-\varepsilon}$ and the error term is $x^{1/4+\theta/2+\varepsilon}d^{3/4-3\theta/2}$ (Proposition 7.1). This allows $d \leq x^\varpi$ for every $\varpi < (1-2\theta)/(5-6\theta)$, and Theorem 1.1 follows. Under Selberg's conjecture the exponent $1/10$ is the one anticipated in [11, Remark 8.1] from square-root cancellation in the modulus aspect. Remark 8.1 discusses the main obstruction to further progress. Duke, Friedlander and Iwaniec [5] have also proved bounds for sums of Weyl sums for quadratic roots that are uniform in the discriminant; Remark 8.3 explains why they do not help here.

1.2. Notation. We write $e(t) = e^{2\pi it}$ and $\hat{\psi}(\xi) = \int_{\mathbb{R}} \psi(t)e(-\xi t) dt$. The letter χ denotes the non-principal character modulo 4, and

$$r(n) = \#\{(x, y) \in \mathbb{Z}^2 : x^2 + y^2 = n\} = 4 \sum_{k|n} \chi(k) \quad (n \geq 1).$$

For a prime ℓ , v_ℓ is the ℓ -adic valuation. The functions $\tau(n)$ and $\omega(n)$ count the divisors and the distinct prime factors of n , and $\mathbf{1}\{\cdot\}$ is an indicator function. Throughout, a and b are fixed, θ is a fixed admissible exponent, and $\varepsilon, \eta > 0$ are small. Implied constants may depend on a, b, ε, η and on the fixed smooth functions F, W and ψ_0 introduced below, but on nothing else unless indicated.

2. THE FAMILIES

Lemma 2.1. *If a and b are both even, then $S_{a,b}(x) \geq S_{a/2,b/2}(x/2)$.*

Proof. If $1 \leq n' \leq x/2$ and $n', n' + a/2, n' + b/2 \in \mathcal{S}$, then $n = 2n' \leq x$ and $n, n + a, n + b \in \mathcal{S}$, because $2\mathcal{S} \subseteq \mathcal{S}$. $\square$

Definition 2.2. A pair $(B, A) \in \mathbb{Z}^2$ is *admissible* if $B > 0$ is odd, $A \notin \{0, B\}$, and neither

$$B \equiv 1, A \equiv 3 \pmod{4} \quad \text{nor} \quad B \equiv 3, A \equiv 2 \pmod{4}.$$

Lemma 2.3. *Let $0 < a < b$ with a or b odd. There are $p_0 \in \{0, a\}$ and an admissible pair (B, A) such that*

$$\{n + p_0, n + p_0 + B, n + p_0 + A\} = \{n, n + a, n + b\} \quad \text{for all } n \in \mathbb{Z}.$$

One such choice is given in Table 1.

case	p_0	B	A	automatic pair	detected
1 a odd, b even, $(a, b) \not\equiv (3, 2) \pmod{4}$	0	a	b	$n, n + a$	$n + b$
2 $(a, b) \equiv (3, 2) \pmod{4}$	a	$b - a$	$-a$	$n + a, n + b$	n
3 a even, b odd, $(a, b) \not\equiv (2, 3) \pmod{4}$	0	b	a	$n, n + b$	$n + a$
4 $(a, b) \equiv (2, 3) \pmod{4}$	a	$b - a$	$-a$	$n + a, n + b$	n
5 a, b odd, $(a, b) \not\equiv (1, 3) \pmod{4}$	0	a	b	$n, n + a$	$n + b$
6 $(a, b) \equiv (1, 3) \pmod{4}$	0	b	a	$n, n + b$	$n + a$

TABLE 1. The choice of (p_0, B, A) in Lemma 2.3.

Proof. The six rows cover all pairs (a, b) with a or b odd. In each row $B > 0$ is odd, and the three numbers $n + p_0, n + p_0 + B, n + p_0 + A$ are $n, n + a, n + b$ in some order, so $A \notin \{0, B\}$. In rows 1, 3 and 5 the excluded congruences are exactly those that would violate admissibility. In rows 2, 4 and 6 we have $(B, A) \equiv (3, 1), (1, 2)$ and $(3, 1) \pmod{4}$ respectively, and these pairs are admissible. $\square$

From now on we assume, as we may by Lemma 2.1, that a or b is odd, and we fix p_0 , B and A as in Lemma 2.3. For $n \in \mathbb{Z}$ we write

$$(2.1) \quad P = n + p_0, \quad T = P + A, \quad Q = 2P + B,$$

so that $\{P, P + B, T\} = \{n, n + a, n + b\}$. The numbers P and $P + B$ form the automatic pair, and T is the detected element. Put

$$(2.2) \quad c = 2A - B, \quad m_d = d^2 + 2cd + B^2 = (d + c)^2 + 4A(B - A).$$

For example, for $\{0, 1, 2\}$ Table 1 gives $p_0 = 0$, $(B, A) = (1, 2)$ and $m_d = d^2 + 6d + 1$.

Lemma 2.4. *Let $d, u, T \in \mathbb{Z}$ satisfy $4dT = u^2 + m_d$, and put $P = T - A$ and $Q = 2P + B$. Then $2dQ = u^2 + d^2 + B^2$ and*

$$4P(P + B) = Q^2 - B^2 = (Q - d)^2 + u^2.$$

Proof. We have $Q = 2T - c$, so $2dQ = 4dT - 2cd = u^2 + m_d - 2cd = u^2 + d^2 + B^2$. Next, $Q - B = 2P$ and $Q + B = 2(P + B)$. Finally, $(Q - d)^2 + u^2 = Q^2 - 2dQ + d^2 + u^2 = Q^2 - B^2$. $\square$

Lemma 2.5. *In Lemma 2.4, suppose that d is a prime with $d \equiv 1 \pmod{4}$ and that $P \geq 1$. Then $P \in \mathcal{S}$ and $P + B \in \mathcal{S}$.*

Proof. Recall that $z \geq 1$ lies in $\mathcal{S}$ if and only if $v_\ell(z)$ is even for every prime $\ell \equiv 3 \pmod{4}$. By Lemma 2.4, $4P(P + B) \in \mathcal{S}$, so $P(P + B) \in \mathcal{S}$. Let $\ell \equiv 3 \pmod{4}$ be prime. If ℓ divides at most one of P and $P + B$, then $v_\ell(P)$ and $v_\ell(P + B)$ are 0 or $v_\ell(P(P + B))$, hence even. If ℓ divides both, then $\ell \mid B$ and $\ell \mid Q = 2P + B$, so $\ell \mid u^2 + d^2$ by Lemma 2.4. Since -1 is not a square modulo ℓ , this forces $\ell \mid d$, which is impossible. $\square$

Lemma 2.6. *Let n be an integer with $P = n + p_0 \geq 1$, and let T be as in (2.1). Then*

$$\#\{(d, u) \in \mathbb{Z}^2 : d \geq 1, 4dT = u^2 + m_d\} \leq r(P(P + B)).$$

Proof. The number $Q = 2P + B$ depends only on n . By Lemma 2.4, the map $(d, u) \mapsto (Q - d, u)$ sends the set on the left injectively into the set of representations of $4P(P + B)$ as a sum of two squares. Finally, $r(4z) = r(z)$. $\square$

3. LOCAL ANALYSIS AT THE PRIME 2

Let $\mathcal{N} \subset \mathbb{Z}_2$ be the set of non-zero 2-adic integers whose odd part is $\equiv 1 \pmod{4}$. Then $\mathcal{S} \setminus \{0\} \subseteq \mathcal{N}$, since an odd element of $\mathcal{S}$ is $\equiv 1 \pmod{4}$. We write $z = 2^{v_2(z)}z'$ with z' odd. Recall that a non-zero $z \in \mathbb{Q}_2$ is a square if and only if $v_2(z)$ is even and $z' \equiv 1 \pmod{8}$.

Lemma 3.1. *Let (B, A) be admissible. There is $n_* \in \mathbb{Z}$ such that*

- (i) $Q_* := 2n_* + B \equiv 1 \pmod{4}$;
- (ii) $n_*(n_* + B)$ is a non-zero square in $\mathbb{Q}_2$;
- (iii) $T_* := n_* + A \in \mathcal{N}$.

Proof. Let $\sigma = 1$ if $B \equiv 1 \pmod{4}$ and $\sigma = -1$ if $B \equiv 3 \pmod{4}$, so that $\sigma B \equiv 1 \pmod{4}$. For an integer $s \geq 1$ and an odd integer o put

$$n = 4^s o \quad (\sigma = 1), \quad n = 4^s o - B \quad (\sigma = -1).$$

Then $Q = 2n + B$ equals $2 \cdot 4^s o + B$, respectively $2 \cdot 4^s o - B$, so $Q \equiv \sigma B \equiv 1 \pmod{4}$ and (i) holds. Moreover $n(n + B) = 4^s o(4^s o + \sigma B)$, and since $o^2 \equiv 1 \pmod{8}$,

$$o(4^s o + \sigma B) \equiv 4^s + \sigma B o \pmod{8}.$$

Hence (ii) holds if and only if $o \equiv c_s \pmod{8}$, where $c_s \equiv \sigma B + 4 \pmod{8}$ if $s = 1$ and $c_s \equiv \sigma B \pmod{8}$ if $s \geq 2$. In both cases $c_s \equiv 1 \pmod{4}$. Finally $T = n + A = 4^s o + C$, where $C = A$ if

$\sigma = 1$ and $C = A - B$ if $\sigma = -1$. We have $C \neq 0$, because $A \notin \{0, B\}$. It remains to choose s , and the class of o modulo a power of 2 compatible with $o \equiv c_s \pmod{8}$, so that $4^s o + C \in \mathcal{N}$.

C odd. We claim that $C \equiv 1 \pmod{4}$. If $\sigma = 1$, then $C = A$ is odd, and admissibility excludes $A \equiv 3 \pmod{4}$. If $\sigma = -1$, then $C = A - B$ is odd, so A is even; admissibility excludes $A \equiv 2 \pmod{4}$, so $A \equiv 0 \pmod{4}$ and $C \equiv -B \equiv 1 \pmod{4}$. We take $s = 2$ and $o = c_2$; then $T \equiv C \equiv 1 \pmod{4}$.

C even. Write $C = 2^t w$ with $t \geq 1$ and w odd.

- If $w \equiv 1 \pmod{4}$, take $s = \max(2, \lceil (t+2)/2 \rceil)$ and $o = c_s$. Then $T = 2^t(w + 2^{2s-t}o)$ with $2s - t \geq 2$, so $w + 2^{2s-t}o \equiv 1 \pmod{4}$.
- If $w \equiv 3 \pmod{4}$ and t is odd, take $s = (t+1)/2$ and $o = c_s$. Then $T = 2^t(w + 2o)$ and $w + 2o \equiv 3 + 2 \equiv 1 \pmod{4}$.
- If $w \equiv 3 \pmod{4}$ and t is even, take $s = t/2 \geq 1$. Then $T = 2^t(w + o)$, and $w + c_s \equiv 0 \pmod{4}$. If $w + c_s \equiv 4 \pmod{8}$, take $o \equiv 4 - w \pmod{16}$; then $o \equiv c_s \pmod{8}$ and $w + o \equiv 4 \pmod{16}$. If $w + c_s \equiv 0 \pmod{8}$, take $o \equiv 8 - w \pmod{32}$; then $o \equiv c_s \pmod{8}$ and $w + o \equiv 8 \pmod{32}$. In both cases $w + o \in \mathcal{N}$.

In every case $T \in \mathcal{N}$. Taking for o a positive integer in the required class gives $n_* \in \mathbb{Z}$ with (i)–(iii). $\square$

Proposition 3.2. *Let (B, A) be admissible and let m_d be as in (2.2). There are integers $v \geq 2$ and $d_0 \equiv 1 \pmod{4}$ with the following property. Put $J = v + 2$ and $j = \lceil (v+2)/2 \rceil$. For all integers $d \equiv d_0 \pmod{2^J}$ and $u \equiv 0 \pmod{2^j}$,*

$$v_2(u^2 + m_d) = v \quad \text{and} \quad 2^{-v}(u^2 + m_d) \equiv d \pmod{4}.$$

Consequently, if in addition $d \mid u^2 + B^2$, then $T = (u^2 + m_d)/(4d)$ is an integer of the form $2^{v-2}T'$ with $T' \equiv 1 \pmod{4}$.

Proof. Let n_* , Q_* and T_* be as in Lemma 3.1. By (ii) there is $t_* \in \mathbb{Z}_2$ with $t_*^2 = 4n_*(n_* + B)$, and $v_2(n_*(n_* + B))$ is even. It is also positive, because B is odd and so exactly one of n_* and $n_* + B$ is even. Hence $v_2(t_*) \geq 2$. Put $d_* = Q_* - t_*$. Then $d_* \equiv Q_* \equiv 1 \pmod{4}$ and

$$d_*(Q_* + t_*) = Q_*^2 - t_*^2 = (2n_* + B)^2 - 4n_*(n_* + B) = B^2.$$

Hence $d_*^2 - 2d_*Q_* + B^2 = d_*(d_* - 2Q_* + Q_* + t_*) = 0$, that is, $2d_*Q_* = d_*^2 + B^2$. This is the relation of Lemma 2.4 with $u = 0$, read 2-adically, and it gives

$$m_{d_*} = d_*^2 + 2cd_* + B^2 = 2d_*(Q_* + c) = 4d_*(n_* + A) = 4d_*T_*.$$

Let $v = v_2(m_{d_*}) = 2 + v_2(T_*) \geq 2$. By (iii), $2^{-v}m_{d_*} = d_*T'_* \equiv d_* \pmod{4}$. Choose $d_0 \in \mathbb{Z}$ with $d_0 \equiv d_* \pmod{2^J}$; then $d_0 \equiv 1 \pmod{4}$. If $d \equiv d_0 \pmod{2^J}$, then $m_d \equiv m_{d_*} \pmod{2^J}$, because m_d is a polynomial in d with integer coefficients. If moreover $u \equiv 0 \pmod{2^j}$, then $u^2 \equiv 0 \pmod{2^J}$ since $2j \geq v + 2 = J$. Hence $u^2 + m_d \equiv m_{d_*} \pmod{2^J}$, and since $v_2(m_{d_*}) = v = J - 2$ we get $v_2(u^2 + m_d) = v$ and $2^{-v}(u^2 + m_d) \equiv 2^{-v}m_{d_*} \equiv d_* \equiv d \pmod{4}$.

For the last assertion note that $m_d \equiv B^2 \pmod{d}$, so $d \mid u^2 + m_d$. Since $d \equiv 1 \pmod{4}$ is odd and $4 \mid u^2 + m_d$, T is an integer with $v_2(T) = v - 2$, and its odd part T' satisfies $dT' = 2^{-v}(u^2 + m_d) \equiv d \pmod{4}$, that is, $T' \equiv 1 \pmod{4}$. $\square$

4. GOOD PRIMES AND THE DEDUCTION OF THEOREM 1.1

We fix v and d_0 as in Proposition 3.2 and put

$$(4.1) \quad J = v + 2, \quad j = \lceil (v+2)/2 \rceil, \quad E = 2^{2j-v} \in \{4, 8\}.$$

Note that $j \leq v$, because $v \geq 2$. Let $d_1 = 8(|c| + B)$. For $d > d_1$ we have $d > B$ and $|2cd + B^2| \leq d^2/2$, so $\frac{1}{2}d^2 \leq m_d \leq 2d^2$.

Definition 4.1. A prime d is *good* if $d \equiv d_0 \pmod{2^J}$, $d > d_1$, and neither m_d nor $m_d/3$ is a perfect square.

Let d be a good prime. Then $d \equiv 1 \pmod{4}$ and $d \nmid B$. Since $m_d \equiv B^2 \pmod{d}$, also $d \nmid m_d$. By Proposition 3.2 with $u = 0$, $v_2(m_d) = v$, so

$$H_d := 2^{-v}m_d$$

is an odd positive integer with $d \nmid H_d$. Since $2^v E = 2^{2j}$, we have

$$(4.2) \quad (2^j \ell)^2 + m_d = 2^v (E \ell^2 + H_d) \quad (\ell \in \mathbb{Z}).$$

Moreover $N_d := E H_d = 4^{j-v} m_d$, and since $j \leq v$, N_d is neither a square nor three times a square: if $N_d = s^2$ or $N_d = 3s^2$, then $m_d = (2^{v-j} s)^2$ or $m_d = 3(2^{v-j} s)^2$.

Fix a smooth $F : \mathbb{R} \rightarrow [0, 1]$ supported in $[1/2, 1]$ and not identically zero, and put $c_F = \int_0^1 F(t) t^{-1/2} dt > 0$. For a good prime d and $x \geq 1$ define

$$(4.3) \quad \Sigma_d(x) = \sum_{\substack{\ell \in \mathbb{Z} \\ d \mid E \ell^2 + H_d}} r(T_d(\ell)) F\left(\frac{T_d(\ell)}{x}\right), \quad T_d(\ell) = \frac{2^{v-2}(E \ell^2 + H_d)}{d} = \frac{(2^j \ell)^2 + m_d}{4d}.$$

By (4.2), and since $m_d \equiv B^2 \pmod{d}$, we have $d \mid E \ell^2 + H_d$ if and only if $d \mid u^2 + B^2$, where $u = 2^j \ell$. Proposition 3.2 therefore shows that for every ℓ in (4.3) the number $T_d(\ell)$ is an integer of the form $2^{v-2} T'$ with $T' \equiv 1 \pmod{4}$.

Proposition 4.2. Let $\mathcal{D}$ be a finite set of good primes and $x \geq x_0$. Then

$$\sum_{d \in \mathcal{D}} \Sigma_d(x) \leq S_{a,b}(x) \cdot \max_{n \leq x} r(T) r(P(P+B)) \ll_{\varepsilon} S_{a,b}(x) x^{\varepsilon},$$

with P and T given by (2.1).

Proof. Let $d \in \mathcal{D}$, and let ℓ be such that the corresponding term in (4.3) is non-zero. Put $u = 2^j \ell$ and $T = T_d(\ell)$. Then T is an integer, $4dT = u^2 + m_d$, $x/2 \leq T \leq x$ and $T \in \mathcal{S}$. Put $P = T - A$ and $n = P - p_0$. Then $\{n, n+a, n+b\} = \{P, P+B, T\}$, and since n is the smallest of these numbers, $T - b \leq n \leq T$. Hence $1 \leq n \leq x$, and $P \geq n \geq 1$, for $x \geq x_0$. By Lemmas 2.4 and 2.5, P and $P+B$ lie in $\mathcal{S}$. So n is counted by $S_{a,b}(x)$. By Lemma 2.6, each such n arises from at most $r(P(P+B))$ pairs (d, u) , and u determines ℓ . The last bound follows from $r(z) \leq 4\tau(z) \ll z^{\varepsilon}$. $\square$

Theorem 4.3. Let $0 < \varpi < \varpi_{\theta} := (1 - 2\theta)/(5 - 6\theta)$. There is $\varepsilon_0 = \varepsilon_0(\varpi, \theta) > 0$ such that for $0 < \varepsilon \leq \varepsilon_0$ and $x \geq x_0(a, b, \varpi, \varepsilon)$,

$$\Sigma_d(x) \gg_{\varepsilon} x^{1/2} d^{-1/2-\varepsilon}$$

for every good prime $d \leq x^{\varpi}$. The implied constant is ineffective.

Deduction of Theorem 1.1. Let $\delta < \delta_{\theta} = \varpi_{\theta}/2$. Choose ϖ with $2\delta < \varpi < \varpi_{\theta}$, and then $\varepsilon \leq \varepsilon_0(\varpi, \theta)$ with $\varpi(\frac{1}{2} - \varepsilon) - 2\varepsilon > \delta$. By Lemma 2.1, applied repeatedly with x replaced by $x/2^i$, we may assume that a or b is odd.

The primes $d \leq x^{\varpi}$ with $d \equiv d_0 \pmod{2^J}$ that are not good are of three kinds. There are $O(1)$ primes $d \leq d_1$. There are $O(1)$ primes with $m_d = s^2$, because $(d+c)^2 - s^2 = -4A(B-A) \neq 0$ has finitely many solutions. Finally, there are the primes with $m_d = 3s^2$. The equation $(d+c)^2 - 3s^2 = -4A(B-A)$ has $O(\log x)$ solutions with $0 < d \leq x$, because its solutions form finitely many orbits under the units of $\mathbb{Z}[\sqrt{3}]$.

Let $\mathcal{D}$ be the set of good primes $d \leq x^{\varpi}$. The prime number theorem in arithmetic progressions gives $\sum_{d \in \mathcal{D}} d^{-1/2-\varepsilon} \gg x^{\varpi(1/2-\varepsilon)} (\log x)^{-1}$. By Theorem 4.3 and Proposition 4.2,

$$S_{a,b}(x) \gg x^{-\varepsilon} \sum_{d \in \mathcal{D}} \Sigma_d(x) \gg x^{1/2+\varpi(1/2-\varepsilon)-\varepsilon} (\log x)^{-1} \gg x^{1/2+\delta}. \quad \square$$

For $\theta = 7/64$ we get $\varpi_\theta = 25/139$ and $\delta_\theta = 25/278$, and for $\theta = 0$ we get $\varpi_\theta = 1/5$ and $\delta_\theta = 1/10$. This proves Corollary 1.2. The rest of the paper proves Theorem 4.3.

5. ARITHMETIC PRELIMINARIES

This section is essentially [11, §5], adapted to the present families. We include the proofs for completeness.

5.1. A smoothed hyperbola identity. Fix a smooth non-increasing $\lambda_0 : \mathbb{R} \rightarrow [0, 1]$ with $\lambda_0(t) = 1$ for $t \leq -\log 2$ and $\lambda_0(t) = 0$ for $t \geq \log 2$. Put $\lambda(t) = \frac{1}{2}(\lambda_0(t) + 1 - \lambda_0(-t))$ and $W(t) = \lambda(\log t)$ for $t > 0$. Then W is smooth with values in $[0, 1]$, and

$$(5.1) \quad W(t) = 1 \quad (t \leq \tfrac{1}{2}), \quad W(t) = 0 \quad (t \geq 2), \quad W(t) + W(1/t) = 1.$$

Lemma 5.1. *Let $s \geq 0$ be an integer and $T = 2^s T'$ with $T' \equiv 1 \pmod{4}$. Then*

$$r(T) = 8 \sum_{k|T} \chi(k) W(2^{s/2} k / \sqrt{T}).$$

Proof. First, $r(T) = r(T') = 4 \sum_{k|T'} \chi(k)$. By (5.1),

$$\sum_{k|T'} \chi(k) = \sum_{k|T'} \chi(k) W(k/\sqrt{T'}) + \sum_{k|T'} \chi(k) W(\sqrt{T'}/k).$$

In the second sum substitute $k = T'/i$ and use $\chi(T'/i) = \chi(i)$, which holds because $T' \equiv 1 \pmod{4}$. So the two sums are equal. Finally, the odd divisors of T are those of T' , χ vanishes on even numbers, and $\sqrt{T'} = 2^{-s/2} \sqrt{T}$. $\square$

5.2. Roots of $\nu^2 \equiv -m_d$. Throughout this subsection and the next, d is a good prime, $m = m_d$ and $H = H_d$. For $n \geq 1$ let

$$\varrho(n) = \#\{\nu \bmod n : \nu^2 + m \equiv 0 \pmod{n}\},$$

a multiplicative function.

Lemma 5.2. (a) *For odd n , $\varrho(n) = \#\{t \bmod n : Et^2 + H \equiv 0 \pmod{n}\}$.*

(b) *Let ℓ be an odd prime and $i \geq 1$. If $\ell \nmid m$ then $\varrho(\ell^i) = 1 + \left(\frac{-m}{\ell}\right)$. In general $\varrho(\ell^i) \leq 2 \ell^{\min(i, v_\ell(m))/2}$. Consequently $\varrho(n) \leq 2^{\omega(n)} \gcd(n, m)^{1/2} \ll_\varepsilon n^\varepsilon d$ for odd n .*

(c) *$\varrho(d^i) = 2$ for all $i \geq 1$.*

Proof. (a) This follows from (4.2), since 2^v and 2^j are invertible modulo n .

(b) For $\ell \nmid m$ use Hensel's lemma. Let $\alpha = v_\ell(m) \geq 1$. For $i \leq \alpha$ the congruence reads $\nu^2 \equiv 0 \pmod{\ell^i}$, which has $\ell^{\lfloor i/2 \rfloor}$ solutions. For $i > \alpha$ a solution has $v_\ell(\nu) = \alpha/2$, so there are none unless α is even. In that case $\nu = \ell^{\alpha/2} \nu_1$, where ν_1 is taken modulo $\ell^{i-\alpha/2}$ and satisfies $\nu_1^2 \equiv -m \ell^{-\alpha} \pmod{\ell^{i-\alpha}}$. There are at most 2 such ν_1 modulo $\ell^{i-\alpha}$, each with $\ell^{\alpha/2}$ lifts. The consequence follows by multiplicativity and $m \leq 2d^2$.

(c) Since $m \equiv B^2 \pmod{d}$ with $d \nmid B$ and $d \equiv 1 \pmod{4}$, we get $\varrho(d^i) = 1 + \left(\frac{-B^2}{d}\right) = 2$. $\square$

For odd k put $f(k) = \varrho(dk)/2$. Then f is multiplicative on odd integers, with

$$(5.2) \quad f(d^i) = 1, \quad f(\ell^i) = \varrho(\ell^i) \quad (\ell \neq d).$$

5.3. The singular series. Let $\psi(n) = \left(\frac{m}{n}\right)$ (Jacobi symbol) for odd $n \geq 1$, and $\psi(n) = 0$ for even n .

Lemma 5.3. (a) ψ is a non-principal real Dirichlet character modulo $8m$.

(b) For $t \geq 1$, $A(t) := \sum_{k \leq t} \chi(k)f(k) \ll_\varepsilon t^{1/2+\varepsilon} d^{1+\varepsilon}$.

(c) The series $\mathfrak{L}_d := \sum_{k \geq 1} \chi(k)f(k)/k$ converges, $|\mathfrak{L}_d| \ll d^{1+\varepsilon}$, and

$$\mathfrak{L}_d = L(1, \chi) L(1, \psi) \mathcal{P}_d \quad \text{with} \quad \mathcal{P}_d \geq \frac{6}{\pi^2} \cdot \frac{\phi(dm)}{dm}.$$

(d) $\mathfrak{L}_d \gg_\varepsilon d^{-\varepsilon}$.

Proof. (a) Write $m = 2^t m'$ with m' odd. For odd $n > 0$, quadratic reciprocity for Jacobi symbols gives

$$\left(\frac{m}{n}\right) = \left(\frac{2}{n}\right)^t (-1)^{\frac{m'-1}{2} \frac{n-1}{2}} \left(\frac{n}{m'}\right),$$

which is a character modulo $8m' \mid 8m$. It is non-principal because m is not a square. If m' is not a square, then $\left(\frac{\cdot}{m'}\right)$ is non-principal, and the Chinese remainder theorem gives $n \equiv 1 \pmod{8}$ with $\psi(n) = \left(\frac{n}{m'}\right) = -1$. If $m' = s^2$, then t is odd. Choose $n \equiv 3 \pmod{8}$ with $n \equiv 1 \pmod{s}$; then $\psi(n) = (-1)^t \cdot (-1)^{\frac{m'-1}{2} \frac{n-1}{2}} \left(\frac{n}{s}\right)^2 = -1$, because t is odd and $m' = s^2 \equiv 1 \pmod{8}$.

(b) We compare the Euler factors of χf and $\chi * \psi$, writing $Y = \ell^{-s}$.

- At $\ell = 2$, both χf and $\chi * \psi$ have factor 1.
- For odd $\ell \nmid dm$, (5.2) and Lemma 5.2(b) give, with $\psi(\ell) = \chi(\ell) \left(\frac{-m}{\ell}\right)$,

$$\sum_{i \geq 0} \chi(\ell^i) f(\ell^i) Y^i = 1 + \left(1 + \left(\frac{-m}{\ell}\right)\right) \frac{\chi(\ell) Y}{1 - \chi(\ell) Y} = \frac{1 - Y^2}{(1 - \chi(\ell) Y)(1 - \psi(\ell) Y)}.$$

- At $\ell = d$, $\chi(d) = 1$ because $d \equiv 1 \pmod{4}$, and $\psi(d) = \left(\frac{B^2}{d}\right) = 1$. The factor of χf is $(1 - Y)^{-1} = (1 - Y) \cdot (1 - Y)^{-2}$.
- For odd $\ell \mid m$, $\psi(\ell) = 0$. The factor of χf is $P_\ell(Y) := \sum_i \chi(\ell)^i \varrho(\ell^i) Y^i = (1 - \chi(\ell) Y) P_\ell(Y) \cdot (1 - \chi(\ell) Y)^{-1}$.

Hence $\chi f = (\chi * \psi) * \beta$, where β is multiplicative, supported on odd integers, and given as follows: $\beta(\ell^2) = -1$ and $\beta(\ell^i) = 0$ for $i \neq 0, 2$ when $\ell \nmid 2dm$; $\beta(d) = -1$ and $\beta(d^i) = 0$ for $i \geq 2$; and $\beta(\ell^i) = \chi(\ell)^i (\varrho(\ell^i) - \varrho(\ell^{i-1}))$ for odd $\ell \mid m$. By Lemma 5.2(b), $|\beta(\ell^i)| \leq 2\ell^{\min(i, \alpha)/2}$ for odd $\ell \mid m$ with $\alpha = v_\ell(m)$, so

$$\sum_{i \geq 0} |\beta(\ell^i)| \ell^{-i(1/2+\varepsilon)} \leq 1 + 2\alpha + 2 \sum_{i \geq 1} \ell^{-i/2} \leq 2\alpha + 5.$$

Therefore $\sum_n |\beta(n)| n^{-1/2-\varepsilon} \leq \zeta(1+2\varepsilon)(1+d^{-1/2}) \prod_{\ell \mid m} (2v_\ell(m) + 5) \ll_\varepsilon d^\varepsilon$.

The Pólya–Vinogradov inequality [2, Ch. 23], applied to the primitive character inducing ψ together with Möbius inversion over the remaining primes dividing $8m$, gives $\Psi^* \ll_\varepsilon m^{1/2+\varepsilon}$, where $\Psi^* := \sup_y |\sum_{n \leq y} \psi(n)|$. The hyperbola method with parameter $\sqrt{y}$ gives $|\sum_{n \leq y} (\chi * \psi)(n)| \leq 3\sqrt{y}(\Psi^* + 1)$ for $y \geq 1$. Finally $A(t) = \sum_{i \leq t} \beta(i) \sum_{n \leq t/i} (\chi * \psi)(n)$, so

$$|A(t)| \ll m^{1/2+\varepsilon} \sum_{i \leq t} |\beta(i)| (t/i)^{1/2} \leq m^{1/2+\varepsilon} t^{1/2+\varepsilon} \sum_i |\beta(i)| i^{-1/2-\varepsilon} \ll t^{1/2+\varepsilon} d^{1+3\varepsilon}.$$

Renaming ε gives (b).

(c) Partial summation and (b) give convergence and $|\mathfrak{L}_d| \leq \int_1^\infty |A(t)| t^{-2} dt \ll d^{1+\varepsilon}$. For $\operatorname{Re} s > 1$ we have $\sum_k \chi(k)f(k)k^{-s} = L(s, \chi)L(s, \psi)\mathcal{P}_d(s)$ with $\mathcal{P}_d(s) = \sum_n \beta(n)n^{-s}$. Both sides

are analytic for $\operatorname{Re} s > 1/2$: the left-hand side because it converges there by (b), and the right-hand side because χ and ψ are non-principal and $\mathcal{P}_d(s)$ converges absolutely. So the identity extends to $s = 1$, with $\mathcal{P}_d := \mathcal{P}_d(1) = \prod_{\ell} \beta_{\ell}(\ell^{-1})$, where $\beta_{\ell}(Y) = \sum_i \beta(\ell^i) Y^i$.

The Euler factors are $1 - \ell^{-2}$ for $\ell \nmid 2dm$, $1 - d^{-1}$ at $\ell = d$, and $(1 - \chi(\ell)\ell^{-1})P_{\ell}(\ell^{-1})$ for odd $\ell \mid m$. In the last case, if $\chi(\ell) = 1$ then $P_{\ell}(\ell^{-1}) \geq 1$. If $\chi(\ell) = -1$, let u be distributed according to Haar measure on $\mathbb{Z}_{\ell}$. Since $\varrho(\ell^i)\ell^{-i} = \operatorname{Prob}(\ell^i \mid u^2 + m)$, we get

$$P_{\ell}(\ell^{-1}) = \sum_{i \geq 0} (-1)^i \operatorname{Prob}(\ell^i \mid u^2 + m) = \operatorname{Prob}(v_{\ell}(u^2 + m) \text{ even}) \geq \operatorname{Prob}(\ell \nmid u^2 + m) = 1 - \frac{1}{\ell},$$

since $\varrho(\ell) = 1$. In all cases the factor is $\geq 1 - \ell^{-1}$, which gives the lower bound for $\mathcal{P}_d$.

(d) We have $L(1, \chi) = \pi/4$. Let ψ^* be the primitive character inducing ψ . Its conductor q^* divides $8m$, and $L(1, \psi) \geq L(1, \psi^*) \phi(8m)/(8m)$. By Siegel's theorem [2, Ch. 21], $L(1, \psi^*) \gg_{\varepsilon} (q^*)^{-\varepsilon}$. Combine these bounds with (c) and with $\phi(n)/n \gg (\log \log 3n)^{-1}$. $\square$

6. A SINGLE-LEVEL TYPE I ESTIMATE

6.1. The automorphic input. Let $G = \operatorname{SL}_2(\mathbb{R})$, acting on the upper half-plane $\mathbb{H}$ by Möbius transformations. For $g = \begin{pmatrix} g_1 & g_2 \\ g_3 & g_4 \end{pmatrix} \in G$ write $g \cdot i = x(g) + iy(g)$, so that

$$y(g) = \frac{1}{g_3^2 + g_4^2}, \quad x(g) = (g_1 g_3 + g_2 g_4) y(g),$$

and for $R > 0$ put

$$u_R(g) = \frac{1}{4}(g_1^2 + (g_2/R)^2 + (g_3 R)^2 + g_4^2 - 2).$$

Since $g_1^2 + g_4^2 + (g_2/R)^2 + (g_3 R)^2 \geq 2|g_1 g_4| + 2|g_2 g_3| \geq 2$, we have $u_R(g) \geq 0$. A short computation gives $u_1(g) = u(g \cdot i, i)$, where $u(z, w) = |z - w|^2 / (4 \operatorname{Im} z \operatorname{Im} w)$ is the standard point-pair invariant [12, Ch. 1]. Hence

$$(6.1) \quad u_1(g_1^{-1} g_2) = u(g_2 \cdot i, g_1 \cdot i) \quad (g_1, g_2 \in G).$$

Fix a Haar measure dg on G . For $q \geq 1$ let $\Gamma = \Gamma_0(q)$. For compactly supported $F : G \rightarrow \mathbb{C}$ put

$$K_q F(g_1, g_2) = \sum_{\gamma \in \Gamma} F(g_1^{-1} \gamma g_2), \quad \Delta_q F(g_1, g_2) = K_q F(g_1, g_2) - \frac{1}{\operatorname{vol}(\Gamma \backslash G)} \int_G F(g) dg.$$

By a *functional* we mean a finitely supported function $\alpha : G \rightarrow \mathbb{R}$. It acts on functions by $\langle f \rangle_{\alpha} = \sum_g \alpha(g) f(g)$, and on functions Φ on $G \times G$ by $\langle \alpha_1 | \Phi | \alpha_2 \rangle = \sum_{g_1, g_2} \alpha_1(g_1) \alpha_2(g_2) \Phi(g_1, g_2)$. We write I for the functional $\langle f \rangle_I = f(\operatorname{id})$.

Theorem 6.1 (Grimmelt–Merikoski [9, Theorem 2.1], [8, Theorem 8.1]). *Let $q \geq 1$, let $\mathbf{X}, \mathbf{Y} > 0$ and $\delta \in (0, 1)$ with $\mathbf{X}/\mathbf{Y} > \delta$. Let f be a smooth function supported on $[-1, 1] \times [1, 2]$ and satisfying $\partial_x^{\nu_1} \partial_y^{\nu_2} f \ll_{\nu_1, \nu_2} \delta^{-\nu_1 - \nu_2}$, and put $F(g) = f(x(g)/\mathbf{X}, y(g)/\mathbf{Y})$. Let α_1, α_2 be functionals. Then for any choice of $Z_0, Z_1, Z_2 \geq 1$ with $Z_0 Z_1 Z_2 \geq \mathbf{X}/\mathbf{Y} + 1$ we have*

$$\langle \alpha_1 | \Delta_q F | \alpha_2 \rangle \ll q^{o(1)} \delta^{-O(1)} \left(\frac{\mathbf{X}}{\mathbf{Y}} \right)^{1/2 + o(1)} Z_0^{\theta(\Gamma_0(q))} \langle \alpha_1 | \Delta_q k_{Z_1^2, \mathbf{X}} | \alpha_1 \rangle^{1/2} \langle \alpha_2 | \Delta_q k_{Z_2^2, 1} | \alpha_2 \rangle^{1/2},$$

for certain smooth functions $k_{Y, R} : G \rightarrow [0, 1]$ which satisfy

$$k_{Y, R}(g) \leq \frac{\mathbf{1}\{u_R(g) \leq Y\}}{\sqrt{1 + u_R(g)}}.$$

This is the case of [9, Theorem 2.1] without Hecke operators: there one takes coefficients $\beta(h)$ supported on $h = 1$, so that the Hecke operator is the identity. The functionals in [9] are general compactly supported linear functionals, which include ours. As in [9], we read $q^{o(1)}(\mathbf{X}/\mathbf{Y})^{o(1)}$ as $\ll_\eta (q\mathbf{X}/\mathbf{Y})^\eta$ for every $\eta > 0$, with implied constants depending on η and on the implied constants in the hypothesis on f . Since $Z_0 \geq 1$, we may replace $\theta(\Gamma_0(q))$ by any admissible θ . We shall apply the theorem with $\delta = 1/(2\Delta)$, where $\Delta \geq 1$ bounds the derivatives of f in the sense that $\partial_x^{\nu_1} \partial_y^{\nu_2} f \ll \Delta^{\nu_1+\nu_2}$; then $\delta^{-O(1)} \ll \Delta^{O(1)}$. Only the pointwise majorant of $k_{Y,R}$ and the polynomial dependence on Δ will be used. The two quantities under the square roots arise in [8] after an application of the Cauchy–Schwarz inequality. We only use upper bounds for their absolute values, and these upper bounds include the contribution of the subtracted integrals.

6.2. Symmetric matrices and Heegner points. For an integer $N \geq 1$ let $\mathcal{Q}_N$ be the set of integral symmetric matrices $\mathfrak{g} = \begin{pmatrix} \mathfrak{a} & \mathfrak{b} \\ \mathfrak{b} & \mathfrak{c} \end{pmatrix}$ with $\mathfrak{a}, \mathfrak{c} > 0$ and $\mathfrak{a}\mathfrak{c} - \mathfrak{b}^2 = N$. The group $\mathrm{SL}_2(\mathbb{Z})$ acts on $\mathcal{Q}_N$ by $\gamma \diamond \mathfrak{g} = \gamma \mathfrak{g} \gamma^t$. To $\mathfrak{g} \in \mathcal{Q}_N$ we attach the point

$$z(\mathfrak{g}) = \frac{\mathfrak{b} + i\sqrt{N}}{\mathfrak{c}} \in \mathbb{H}.$$

Every positive definite real symmetric matrix of determinant 1 has the form hh^t with $h \in G$, and $h \cdot i$ depends only on hh^t : if $h = \begin{pmatrix} y^{1/2} & xy^{-1/2} \\ 0 & y^{-1/2} \end{pmatrix} k$ with $k \in \mathrm{SO}(2)$, then $hh^t = \begin{pmatrix} y+x^2/y & x/y \\ x/y & 1/y \end{pmatrix}$ and $h \cdot i = x + iy$. Hence if $\mathfrak{g} = N^{1/2}hh^t$, then

$$(6.2) \quad z(\mathfrak{g}) = h \cdot i, \quad \gamma \diamond \mathfrak{g} = N^{1/2}(\gamma h)(\gamma h)^t, \quad z(\gamma \diamond \mathfrak{g}) = \gamma \cdot z(\mathfrak{g}) \quad (\gamma \in \mathrm{SL}_2(\mathbb{Z})).$$

We call $\mathfrak{g}$ *reduced* if $|2\mathfrak{b}| \leq \mathfrak{c} \leq \mathfrak{a}$. This means that $z(\mathfrak{g})$ lies in the closure of the standard fundamental domain, since $|z(\mathfrak{g})|^2 = \mathfrak{a}/\mathfrak{c}$; in particular $\mathrm{Im} z(\mathfrak{g}) \geq \sqrt{3}/2$. Every $\mathrm{SL}_2(\mathbb{Z})$ -orbit in $\mathcal{Q}_N$ contains a reduced element, and there are finitely many reduced elements.

Lemma 6.2. *Let $N \geq 1$ be neither a square nor three times a square.*

- (a) *The stabiliser of every $\mathfrak{g} \in \mathcal{Q}_N$ in $\mathrm{SL}_2(\mathbb{Z})$ is $\{\pm I\}$.*
- (b) *Let Λ_N be a set of reduced representatives of the $\mathrm{SL}_2(\mathbb{Z})$ -orbits on $\mathcal{Q}_N$, and for $\mathfrak{z} \in \Lambda_N$ fix $\sigma_{\mathfrak{z}} \in G$ with $N^{1/2}\sigma_{\mathfrak{z}}\sigma_{\mathfrak{z}}^t = \mathfrak{z}$. Let $q \geq 1$, let $\mathcal{T}_q$ be a set of representatives of $\Gamma_0(q) \backslash \mathrm{SL}_2(\mathbb{Z})$, and let $\mathcal{Q} \subseteq \mathcal{Q}_N$ be invariant under $\Gamma_0(q)$. Define the functional $\alpha_{\mathcal{Q}}$ by*

$$\langle f \rangle_{\alpha_{\mathcal{Q}}} = \frac{1}{2} \sum_{\mathfrak{z} \in \Lambda_N} \sum_{\substack{\tau \in \mathcal{T}_q \\ \tau \diamond \mathfrak{z} \in \mathcal{Q}}} f(\tau \sigma_{\mathfrak{z}}).$$

Then for every compactly supported $F : G \rightarrow \mathbb{C}$ of the form $F(g) = F_{\diamond}(gg^t)$,

$$\langle I | K_q F | \alpha_{\mathcal{Q}} \rangle = \sum_{\mathfrak{g} \in \mathcal{Q}} F_{\diamond}(N^{-1/2}\mathfrak{g}).$$

Proof. (a) For $w \in \mathbb{Z}^2$ we have $(\gamma^t w)^t \mathfrak{g} (\gamma^t w) = w^t (\gamma \diamond \mathfrak{g}) w$. So $\gamma \diamond \mathfrak{g} = \mathfrak{g}$ says that γ^t is a proper automorph of the positive definite binary form $\mathfrak{a}X^2 + 2\mathfrak{b}XY + \mathfrak{c}Y^2$. A positive definite form with a proper automorph other than $\pm I$ is equivalent to a multiple of $X^2 + Y^2$ or of $X^2 + XY + Y^2$. The only such forms with even middle coefficient are equivalent to $[t, 0, t]$ or $[2t, 2t, 2t]$, of determinants t^2 and $3t^2$, which are excluded.

(b) Write $\Gamma = \Gamma_0(q)$. For $\tau \in \mathrm{SL}_2(\mathbb{Z})$ and $\gamma \in \Gamma$, (6.2) gives $F(\gamma \tau \sigma_{\mathfrak{z}}) = F_{\diamond}(N^{-1/2} \gamma \tau \diamond \mathfrak{z})$. As γ runs over Γ , the matrix $\gamma \tau \diamond \mathfrak{z}$ runs over the Γ -orbit of $\tau \diamond \mathfrak{z}$, and by (a) each element of this orbit is attained exactly twice. Every element $g \diamond \mathfrak{z}$ of the $\mathrm{SL}_2(\mathbb{Z})$ -orbit of $\mathfrak{z}$ lies in such an orbit: write $g = \gamma \tau$ with $\tau \in \mathcal{T}_q$. These orbits are pairwise distinct: if $\tau' \diamond \mathfrak{z} = \gamma \tau \diamond \mathfrak{z}$ with $\tau, \tau' \in \mathcal{T}_q$, then

$\tau'^{-1}\gamma\tau = \pm I$ by (a), so $\Gamma\tau' = \Gamma\tau$, as $-I \in \Gamma$. Since $\mathcal{Q}$ is a union of Γ -orbits, we obtain

$$\langle I | K_q F | \alpha_{\mathcal{Q}} \rangle = \frac{1}{2} \sum_{\mathfrak{z} \in \Lambda_N} \sum_{\substack{\tau \in \mathcal{T}_q \\ \tau \circ \mathfrak{z} \in \mathcal{Q}}} \sum_{\gamma \in \Gamma} F(\gamma\tau\sigma_{\mathfrak{z}}) = \sum_{\mathfrak{g} \in \mathcal{Q}} F_{\diamond}(N^{-1/2}\mathfrak{g}).$$

All sums are finite because F has compact support. $\square$

6.3. Two kernel bounds at a single level.

Lemma 6.3. *Let $q \geq 2$, $Y \geq 1$ and $R > 0$. Then*

$$\sum_{\gamma \in \Gamma_0(q)} \frac{\mathbf{1}\{u_R(\gamma) \leq Y\}}{\sqrt{1 + u_R(\gamma)}} \ll_{\eta} \left(1 + R + \frac{Y^{1/2} + R^{-1}}{q}\right) (Y + R^{-1})^{\eta}$$

and

$$\frac{1}{\text{vol}(\Gamma_0(q) \backslash G)} \int_G \frac{\mathbf{1}\{u_R(g) \leq Y\}}{\sqrt{1 + u_R(g)}} dg \ll \frac{Y^{1/2}}{q}.$$

Proof. Let $\gamma = \begin{pmatrix} \gamma_1 & \gamma_2 \\ \gamma_3 & \gamma_4 \end{pmatrix} \in \Gamma_0(q)$ with $u_R(\gamma) \leq Y$. Then $\gamma_1^2 + \gamma_4^2 + (\gamma_2/R)^2 + (\gamma_3R)^2 \leq 4Y + 2 \leq 6Y$, and

$$1 + u_R(\gamma) \geq \frac{1}{4} \max(2, \gamma_1^2, \gamma_4^2, (\gamma_3R)^2).$$

The terms with $\gamma_3 = 0$. Here $\gamma_1\gamma_4 = 1$, so $\gamma_1 = \gamma_4 = \pm 1$ and $u_R(\gamma) = \gamma_2^2/(4R^2)$. These terms contribute at most

$$2 \sum_{|b| \leq 2R\sqrt{Y}} \left(1 + \frac{b^2}{4R^2}\right)^{-1/2} \leq 2 + 4 \sum_{1 \leq b \leq 2R\sqrt{Y}} \min\left(1, \frac{2R}{b}\right) \ll 1 + R \log(2 + Y).$$

The terms with $\gamma_3 \neq 0$. Then $q \mid \gamma_3$, so $|\gamma_3| \geq 2$, and $|\gamma_3| \leq \sqrt{6Y}/R$. Fix such a γ_3 . Let $M = \max(|\gamma_1|, |\gamma_4|)$, so that $1 \leq M \leq \sqrt{6Y}$; $M = 0$ is impossible because $\gamma_2\gamma_3 = -1$ has no solutions. We count the triples $(\gamma_1, \gamma_2, \gamma_4)$ with $\gamma_1\gamma_4 - \gamma_2\gamma_3 = 1$ and $A/2 < M \leq A$, for a power of two A . Then $|\gamma_2\gamma_3| = |\gamma_1\gamma_4 - 1| \leq A^2 + 1$, so γ_2 takes at most $1 + 2(A^2 + 1)/|\gamma_3|$ values. For each γ_2 , the number $1 + \gamma_2\gamma_3$ is non-zero because $|\gamma_3| \geq 2$, and the number of pairs (γ_1, γ_4) with $\gamma_1\gamma_4 = 1 + \gamma_2\gamma_3$ is $2\tau(|1 + \gamma_2\gamma_3|) \ll_{\eta} A^{2\eta}$. The weight is at most $2/\max(M, |\gamma_3|R) \leq 4/\max(A, |\gamma_3|R)$. Summing over $A \leq 2\sqrt{6Y}$, the contribution of γ_3 is

$$\ll_{\eta} Y^{\eta} \sum_A \left(1 + \frac{A^2}{|\gamma_3|}\right) \frac{1}{\max(A, |\gamma_3|R)} \ll Y^{\eta} \left(\frac{\log(2 + Y)}{\max(1, |\gamma_3|R)} + \frac{Y^{1/2}}{|\gamma_3|} \right).$$

Now sum over $\gamma_3 = qc'$ with $1 \leq |c'| \leq C'$, where $C' = \sqrt{6Y}/(qR)$. We have

$$\sum_{c'} \frac{Y^{1/2}}{q|c'|} \ll \frac{Y^{1/2}}{q} \log(2 + C'), \quad \sum_{c'} \frac{1}{\max(1, q|c'|R)} \ll \frac{\log(2 + C')}{qR}.$$

For the second sum, if $qR \geq 1$ this is the harmonic sum, and if $qR < 1$ the terms with $|c'| \leq (qR)^{-1}$ contribute at most $2(qR)^{-1}$. Since $\log(2 + C') \ll \log(2 + Y + R^{-1})$, the terms with $\gamma_3 \neq 0$ contribute $\ll_{\eta} q^{-1}(Y^{1/2} + R^{-1})(Y + R^{-1})^{2\eta}$. Renaming η gives the first bound.

For the second bound, let $D_R = \text{diag}(R^{1/2}, R^{-1/2})$. Then $u_R(g) = u_1(D_R^{-1}gD_R)$, and Haar measure is invariant under conjugation. In the Cartan decomposition $g = k_1 \text{diag}(e^{\rho/2}, e^{-\rho/2}) k_2$ we have $u_1(g) = \sinh^2(\rho/2)$, and dg is a constant multiple of $\sinh \rho d\rho dk_1 dk_2$. Since $du_1 = \frac{1}{2} \sinh \rho d\rho$, the integral is $\ll \int_0^Y (1 + u)^{-1/2} du \ll Y^{1/2}$. Finally, $\text{vol}(\Gamma_0(q) \backslash G) = [\text{SL}_2(\mathbb{Z}) : \Gamma_0(q)] \text{vol}(\text{SL}_2(\mathbb{Z}) \backslash G) \gg q$. $\square$

Lemma 6.4. *Let $E \in \{4, 8\}$, let $H \geq 1$ be odd, and let $d \geq 3$ be a prime with $d \nmid H$. Suppose that $N = EH$ is neither a square nor three times a square, and put $q = 4Ed$. Let $\mathcal{Q} \subseteq \{\mathfrak{g} \in \mathcal{Q}_N : d \mid \mathfrak{c}\}$ be invariant under $\Gamma_0(q)$, and let $\alpha = \alpha_{\mathcal{Q}}$ be as in Lemma 6.2. Then, for every function $k : G \rightarrow [0, 1]$ with $k(g) \leq \mathbf{1}\{u_1(g) \leq 1\}$,*

$$0 \leq \langle \alpha | K_q k | \alpha \rangle \ll_{\eta} N^{1+\eta} \quad \text{and} \quad \frac{\langle 1 \rangle_{\alpha}^2}{\text{vol}(\Gamma_0(q) \backslash G)} \int_G k(g) dg \ll_{\eta} N^{1+\eta}.$$

Proof. For $\mathfrak{g} = \begin{pmatrix} \mathfrak{a} & \mathfrak{b} \\ \mathfrak{b} & \mathfrak{c} \end{pmatrix}$ let $\varphi_{\mathfrak{g}}(X, Y) = \mathfrak{a}X^2 + 2\mathfrak{b}XY + \mathfrak{c}Y^2$, and for $\mathfrak{z} \in \Lambda_N$ write $\sigma_1 = \sigma_{\mathfrak{z}}$ and $z = z(\mathfrak{z}) = \sigma_{\mathfrak{z}} \cdot i$.

Step 1: the number of τ . We claim that for every $\mathfrak{g} \in \mathcal{Q}_N$,

$$(6.3) \quad \#\{\tau \in \mathcal{T}_q : \tau \diamond \mathfrak{g} \in \mathcal{Q}\} \leq 12E \leq 96.$$

Let $\tau, \tau' \in \text{SL}_2(\mathbb{Z})$ have bottom rows (c_0, d_0) and (c'_0, d'_0) with $(c'_0, d'_0) \equiv \lambda(c_0, d_0) \pmod{q}$ for some integer λ . Then the bottom row of $\tau'\tau^{-1}$ is $\equiv (0, \lambda) \pmod{q}$, so $\Gamma_0(q)\tau' = \Gamma_0(q)\tau$. Hence the map sending $\tau \in \mathcal{T}_q$ to the point $(c_0 : d_0) \in \mathbb{P}^1(\mathbb{Z}/q\mathbb{Z})$ given by its bottom row is injective. By the Chinese remainder theorem, $\mathbb{P}^1(\mathbb{Z}/q\mathbb{Z}) = \mathbb{P}^1(\mathbb{Z}/4E\mathbb{Z}) \times \mathbb{P}^1(\mathbb{F}_d)$. The lower right entry of $\tau \diamond \mathfrak{g}$ is $\varphi_{\mathfrak{g}}(c_0, d_0)$, and $\tau \diamond \mathfrak{g} \in \mathcal{Q}$ requires $d \mid \varphi_{\mathfrak{g}}(c_0, d_0)$, which depends only on the image of $(c_0 : d_0)$ in $\mathbb{P}^1(\mathbb{F}_d)$. The form $\varphi_{\mathfrak{g}}$ does not vanish identically modulo d : otherwise $d^2 \mid \mathfrak{a}\mathfrak{c} - \mathfrak{b}^2 = N$, which is impossible because d is odd and $d \nmid H$. So it has at most two zeros in $\mathbb{P}^1(\mathbb{F}_d)$. Since $\#\mathbb{P}^1(\mathbb{Z}/2^r\mathbb{Z}) = 3 \cdot 2^{r-1}$, this gives (6.3).

Step 2: reduction to a count of pairs. By definition,

$$\langle \alpha | K_q k | \alpha \rangle = \frac{1}{4} \sum_{\mathfrak{z}_1, \mathfrak{z}_2 \in \Lambda_N} \sum_{\substack{\tau_1, \tau_2 \in \mathcal{T}_q \\ \tau_i \diamond \mathfrak{z}_i \in \mathcal{Q}}} \sum_{\gamma \in \Gamma_0(q)} k(\sigma_{\mathfrak{z}_1}^{-1} \tau_1^{-1} \gamma \tau_2 \sigma_{\mathfrak{z}_2}),$$

which is non-negative. By (6.1), the summand is at most $\mathbf{1}\{u(\gamma\tau_2 \cdot z_2, \tau_1 \cdot z_1) \leq 1\}$, where $z_i = z(\mathfrak{z}_i)$. Fix τ_1 . The map $(\gamma, \tau_2) \mapsto g = \tau_1^{-1} \gamma \tau_2$ is a bijection from $\Gamma_0(q) \times \mathcal{T}_q$ onto $\text{SL}_2(\mathbb{Z})$. Under it, $u(\gamma\tau_2 \cdot z_2, \tau_1 \cdot z_1) = u(g \cdot z_2, z_1)$, and $\tau_2 \diamond \mathfrak{z}_2 \in \mathcal{Q}$ if and only if $\gamma\tau_2 \diamond \mathfrak{z}_2 = \tau_1 g \diamond \mathfrak{z}_2 \in \mathcal{Q}$. Therefore, by (6.3),

$$\langle \alpha | K_q k | \alpha \rangle \leq \frac{96}{4} \sum_{\mathfrak{z}_1, \mathfrak{z}_2 \in \Lambda_N} \sum_{g \in \text{SL}_2(\mathbb{Z})} \mathbf{1}\{u(g \cdot z_2, z_1) \leq 1\} = 48 \sum_{\mathfrak{z}_1 \in \Lambda_N} \#\{\mathfrak{w} \in \mathcal{Q}_N : u(z(\mathfrak{w}), z(\mathfrak{z}_1)) \leq 1\}.$$

Here we used (6.2) and Lemma 6.2(a): as $\mathfrak{z}_2$ runs over Λ_N and g over $\text{SL}_2(\mathbb{Z})$, the matrix $\mathfrak{w} = g \diamond \mathfrak{z}_2$ runs over $\mathcal{Q}_N$, taking each value twice, and $z(\mathfrak{w}) = g \cdot z_2$.

Step 3: counting. Let $\varrho_N(\mathfrak{c}) = \#\{\beta \bmod \mathfrak{c} : \beta^2 + N \equiv 0 \pmod{\mathfrak{c}}\}$. As in Lemma 5.2(b), $\varrho_N(\ell^i) \leq 2\ell^{\min(i, v_{\ell}(N))/2}$ for odd primes ℓ . Similarly $\varrho_N(2^i) \leq 4 \cdot 2^{\min(i, v_2(N))/2}$. Indeed, for $i \leq v_2(N)$ the solutions are the multiples of $2^{\lceil i/2 \rceil}$. For $i > v_2(N) =: \alpha$ a solution has the form $\beta = 2^{\alpha/2} \beta_1$, where β_1 is taken modulo $2^{i-\alpha/2}$ and $\beta_1^2 \equiv -N2^{-\alpha} \pmod{2^{i-\alpha}}$; there are at most 4 such β_1 modulo $2^{i-\alpha}$, each with $2^{\alpha/2}$ lifts. Hence $\varrho_N(\mathfrak{c}) \leq 4 \cdot 2^{\omega(\mathfrak{c})} \gcd(\mathfrak{c}, N)^{1/2}$, and since $2^{\omega(i\mathfrak{c}')} \leq \tau(i)\tau(\mathfrak{c}')$,

$$\sum_{\mathfrak{c} \leq y} \varrho_N(\mathfrak{c}) \leq 4 \sum_{i|N} i^{1/2} \tau(i) \sum_{\mathfrak{c}' \leq y/i} \tau(\mathfrak{c}') \ll y \log(2y) \tau(N)^2 \ll_{\eta} y N^{\eta} \log(2y).$$

A reduced $\mathfrak{z}$ satisfies $\mathfrak{c}^2 \leq \mathfrak{a}\mathfrak{c} = N + \mathfrak{b}^2 \leq N + \mathfrak{c}^2/4$, so $\mathfrak{c} \leq 2\sqrt{N/3}$, and $\mathfrak{b}$ lies in an interval of length $\mathfrak{c}$; hence $\#\Lambda_N \leq 2 \sum_{\mathfrak{c} \leq 2\sqrt{N}} \varrho_N(\mathfrak{c}) \ll N^{1/2+2\eta}$.

Now fix $\mathfrak{z}_1 \in \Lambda_N$ with $z_1 = x_1 + iy_1$, so that $y_1 \geq \sqrt{3}/2$, and let $\mathfrak{w} \in \mathcal{Q}_N$ with $w = z(\mathfrak{w}) = x_w + iy_w$ and $u(w, z_1) \leq 1$, that is $|w - z_1|^2 \leq 4y_w y_1$. Then $(y_w - y_1)^2 \leq 4y_w y_1$, so $y_w/y_1 \in [3 - 2\sqrt{2}, 3 + 2\sqrt{2}] \subset [1/6, 6]$. Also $(x_w - x_1)^2 \leq 4y_1^2$. Since $y_w = \sqrt{N}/\mathfrak{c}$ and $x_w = \mathfrak{b}/\mathfrak{c}$, we

get $\mathfrak{c} \leq 6\sqrt{N}/y_1 < 7\sqrt{N}$, and $\mathfrak{b}$ lies in an interval of length $10\mathfrak{c}y_1 \leq 60\sqrt{N}$. Given $\mathfrak{b}$ and $\mathfrak{c}$, $\mathfrak{a}$ is determined. Therefore

$$\#\{\mathfrak{w} : u(z(\mathfrak{w}), z_1) \leq 1\} \leq \sum_{\mathfrak{c} \leq 7\sqrt{N}} \left(\frac{60\sqrt{N}}{\mathfrak{c}} + 1 \right) \varrho_N(\mathfrak{c}) \ll_{\eta} N^{1/2+2\eta},$$

by partial summation. Together with the bound for $\#\Lambda_N$, this proves the first assertion after renaming η .

For the second assertion, $\langle 1 \rangle_{\alpha} \leq \frac{1}{2} \cdot 96 \cdot \#\Lambda_N \ll N^{1/2+2\eta}$ by (6.3), and $\text{vol}(\Gamma_0(q) \backslash G)^{-1} \int_G k \ll q^{-1}$ by Lemma 6.3. $\square$

6.4. The Type I estimate.

Proposition 6.5. *Let $E \in \{4, 8\}$, let $H \geq 1$ be odd, and let $d \geq 3$ be a prime with $d \nmid H$. Suppose that $N = EH$ is neither a square nor three times a square, and put $q = 4Ed$. Let $\kappa \in \{1, 3\}$, $\Delta \geq 1$, $X \geq N^{1/2}$ and $1 \leq K \leq qX$. Let $\psi_1, \psi_2 : \mathbb{R} \rightarrow \mathbb{C}$ be smooth, with ψ_1 supported in $[1, 2]$ and ψ_2 supported in $[-1, 1]$, and suppose that $\|\psi_i^{(\nu)}\|_{\infty} \leq C_{\nu} \Delta^{\nu}$ for all $\nu \geq 0$. Put*

$$\varrho_{E,H}(\mu) = \#\{\ell \bmod \mu : E\ell^2 + H \equiv 0 \pmod{\mu}\}$$

and

$$\Xi = \sum_{\substack{\mu \equiv \kappa d \pmod{4d}}} \psi_1\left(\frac{\mu}{K}\right) \left(\sum_{\substack{\ell \in \mathbb{Z} \\ \mu | E\ell^2 + H}} \psi_2\left(\frac{\ell}{X}\right) - \frac{\varrho_{E,H}(\mu)}{\mu} X \int_{\mathbb{R}} \psi_2(t) dt \right).$$

Then for every $\eta > 0$ there is $C(\eta)$ such that

$$\Xi \ll_{\eta} \Delta^{C(\eta)} (qXH)^{\eta} X^{1/2} H^{1/4} \left(1 + \frac{X}{dH^{1/2}}\right)^{\theta},$$

where the implied constant depends only on η and on the C_{ν} .

Proof. We may assume that $\eta < 1/4$. Write $\Gamma = \Gamma_0(q)$, $\varrho = \varrho_{E,H}$, and

$$\mathfrak{M} = \sum_{\mu \equiv \kappa d \pmod{4d}} \psi_1\left(\frac{\mu}{K}\right) \frac{\varrho(\mu)}{\mu} X \int_{\mathbb{R}} \psi_2.$$

Step 1: Poisson summation. Let $Y \geq 1$ and let ψ be smooth and supported in $[-1, 1]$, with $\|\psi^{(\nu)}\|_{\infty} \leq C_{\nu} \Delta^{\nu}$. Splitting ℓ into residue classes modulo μ and applying Poisson summation, we get

$$(6.4) \quad \sum_{\substack{\ell \in \mathbb{Z} \\ \mu | E\ell^2 + H}} \psi\left(\frac{\ell}{Y}\right) = \frac{Y}{\mu} \sum_{h \in \mathbb{Z}} \hat{\psi}\left(\frac{hY}{\mu}\right) \sum_{\substack{\nu \bmod \mu \\ E\nu^2 + H \equiv 0 \pmod{\mu}}} e\left(\frac{h\nu}{\mu}\right).$$

The term $h = 0$ equals $(Y/\mu)\varrho(\mu)\hat{\psi}(0)$. For $\xi \neq 0$, integration by parts gives the bound $|\hat{\psi}(\xi)| \leq 2C_A \Delta^A (2\pi|\xi|)^{-A}$. Suppose that $K \leq Y^{1-\eta}$. For $\mu \leq 2K$ and $h \neq 0$ we then have $|hY/\mu| \geq |h|Y^{\eta}/2$. Using $\varrho(\mu) \leq \mu$, we see that the terms with $h \neq 0$ contribute, summed over $\mu \leq 2K$, at most

$$\ll_A \Delta^A \sum_{\mu \leq 2K} Y Y^{-\eta A} \ll \Delta^A Y^{2-\eta A} \leq \Delta^A Y^{-1} \quad (A = \lceil 3/\eta \rceil).$$

If $K \leq X^{1-\eta}$, we apply this with $Y = X$ and $\psi = \psi_2$. The terms $h = 0$ give exactly $\mathfrak{M}$, so $\Xi \ll \Delta^{\lceil 3/\eta \rceil}$, and we are done. From now on let $K > X^{1-\eta}$. Put $X_1 = X$ and $X_2 = K^{1/(1-\eta)}$, so

that $X \leq X_2 \leq (qX)^2$, and for $i \in \{1, 2\}$ let

$$\Xi_i = \sum_{\mu \equiv \kappa d \pmod{4d}} \psi_1\left(\frac{\mu}{K}\right) \sum_{\substack{\ell \in \mathbb{Z} \\ \mu | E\ell^2 + H}} \frac{X}{X_i} \psi_2\left(\frac{\ell}{X_i}\right).$$

Then $\Xi = \Xi_1 - \Xi_2 + (\Xi_2 - \mathfrak{M})$. Since $K = X_2^{1-\eta}$, Step 1 with $Y = X_2$ and $\psi = (X/X_2)\psi_2$ gives $\Xi_2 - \mathfrak{M} \ll \Delta^{\lceil 3/\eta \rceil}$.

Step 2: passage to symmetric matrices. Let

$$\mathcal{Q}^\kappa = \{\mathfrak{g} \in \mathcal{Q}_N : \mathfrak{b} \equiv 0 \pmod{E}, \mathfrak{c} \equiv \kappa Ed \pmod{4Ed}\}.$$

The map $(\mu, \ell) \mapsto \begin{pmatrix} (E\ell^2 + H)/\mu & E\ell \\ E\ell & E\mu \end{pmatrix}$ is a bijection from the set of pairs $(\mu, \ell) \in \mathbb{Z}^2$ with $\mu \geq 1$, $\mu \equiv \kappa d \pmod{4d}$ and $\mu \mid E\ell^2 + H$ onto $\mathcal{Q}^\kappa$. Indeed, the determinant of the image is $E(E\ell^2 + H) - E^2\ell^2 = N$. Conversely, for $\mathfrak{g} \in \mathcal{Q}^\kappa$ write $\mathfrak{b} = E\ell$ and $\mathfrak{c} = E\mu$; then $\mu \equiv \kappa d \pmod{4d}$ and $\mathfrak{a}\mu = E\ell^2 + H$.

The set $\mathcal{Q}^\kappa$ is Γ -invariant. Let $\gamma = \begin{pmatrix} \gamma_1 & \gamma_2 \\ \gamma_3 & \gamma_4 \end{pmatrix} \in \Gamma$, so that $q \mid \gamma_3$ and γ_4 is odd. The lower right entry of $\gamma \diamond \mathfrak{g}$ is $\gamma_3^2 \mathfrak{a} + 2\gamma_3 \gamma_4 \mathfrak{b} + \gamma_4^2 \mathfrak{c} \equiv \gamma_4^2 \mathfrak{c} \pmod{q}$. Moreover $\gamma_4^2 \mathfrak{c} \equiv \mathfrak{c} \pmod{4Ed}$, because $4 \mid \gamma_4^2 - 1$ and $Ed \mid \mathfrak{c}$. The off-diagonal entry is $\gamma_1 \gamma_3 \mathfrak{a} + (\gamma_1 \gamma_4 + \gamma_2 \gamma_3) \mathfrak{b} + \gamma_2 \gamma_4 \mathfrak{c} \equiv 0 \pmod{E}$, because E divides γ_3 , $\mathfrak{b}$ and $\mathfrak{c}$.

By hypothesis, N is neither a square nor three times a square, so Lemma 6.2 applies. Let $\alpha = \alpha_{\mathcal{Q}^\kappa}$. For a real symmetric matrix $\mathfrak{h} = \begin{pmatrix} h_{11} & h_{12} \\ h_{12} & h_{22} \end{pmatrix}$ and $i \in \{1, 2\}$ put

$$F_{\diamond, i}(\mathfrak{h}) = \psi_1\left(\frac{h_{22}N^{1/2}}{EK}\right) \frac{X}{X_i} \psi_2\left(\frac{h_{12}N^{1/2}}{EX_i}\right), \quad F_i(g) = F_{\diamond, i}(gg^t).$$

If $\mathfrak{g} \in \mathcal{Q}^\kappa$ corresponds to (μ, ℓ) , then $F_{\diamond, i}(N^{-1/2}\mathfrak{g}) = \psi_1(\mu/K)(X/X_i)\psi_2(\ell/X_i)$. Hence Lemma 6.2 gives

$$\Xi_i = \langle I | K_q F_i | \alpha \rangle \quad (i = 1, 2).$$

Since $gg^t = \begin{pmatrix} * & x(g)/y(g) \\ x(g)/y(g) & 1/y(g) \end{pmatrix}$, we have $F_i(g) = f_i(x(g)/\mathbf{X}_i, y(g)/\mathbf{Y})$ with

$$\mathbf{Y} = \frac{N^{1/2}}{2EK}, \quad \mathbf{X}_i = \frac{X_i}{K}, \quad f_i(x', y') = \frac{X}{X_i} \psi_1\left(\frac{2}{y'}\right) \psi_2\left(\frac{2x'}{y'}\right).$$

The function f_i is supported in $[-1, 1] \times [1, 2]$, and $\partial_{x'}^{\nu_1} \partial_{y'}^{\nu_2} f_i \ll \Delta^{\nu_1 + \nu_2}$ with implied constants depending only on the C_ν . Moreover $\mathbf{X}_i/\mathbf{Y} = 2EX_i/N^{1/2} \geq 2E > 1$. Also, F_i has compact support, because it is right $\text{SO}(2)$ -invariant and vanishes unless $g \cdot i$ lies in a fixed compact set.

In the Iwasawa coordinates $g = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y^{1/2} & 0 \\ 0 & y^{-1/2} \end{pmatrix} k$, $k \in \text{SO}(2)$, Haar measure is a constant multiple of $y^{-2} dx dy dk$. Hence, for some constant $c_G > 0$,

$$\int_G F_i(g) dg = c_G \frac{\mathbf{X}_i}{\mathbf{Y}} \iint f_i(x', y') \frac{dx' dy'}{y'^2} = c_G \frac{X}{K\mathbf{Y}} \iint \psi_1\left(\frac{2}{y'}\right) \psi_2\left(\frac{2x'}{y'}\right) \frac{dx' dy'}{y'^2},$$

which does not depend on i . Therefore

$$(6.5) \quad \Xi_1 - \Xi_2 = \langle I | \Delta_q F_1 | \alpha \rangle - \langle I | \Delta_q F_2 | \alpha \rangle.$$

Step 3: application of Theorem 6.1. We apply Theorem 6.1 to F_1 and to $(X_2/X)F_2$, with $\delta = 1/(2\Delta)$, $Z_1 = q$, $Z_2 = 1$ and $Z_0 = \max\{1, (\mathbf{X}_i/\mathbf{Y} + 1)/q\}$. By Lemma 6.4, applied to $\mathcal{Q}^\kappa \subseteq \{\mathfrak{g} : d \mid \mathfrak{c}\}$ and $k = k_{1,1}$,

$$|\langle \alpha | \Delta_q k_{1,1} | \alpha \rangle| \leq \langle \alpha | K_q k_{1,1} | \alpha \rangle + \frac{\langle 1 \rangle_\alpha^2}{\text{vol}(\Gamma \backslash G)} \int_G k_{1,1} \ll N^{1+\eta}.$$

By Lemma 6.3 with $Y = q^2$ and $R = \mathbf{X}_i$,

$$|\langle I | \Delta_q k_{q^2, \mathbf{X}_i} | I \rangle| \leq \sum_{\gamma \in \Gamma} k_{q^2, \mathbf{X}_i}(\gamma) + \frac{1}{\text{vol}(\Gamma \backslash G)} \int_G k_{q^2, \mathbf{X}_i} \ll \left(1 + \mathbf{X}_i + 1 + \frac{1}{q\mathbf{X}_i}\right) (q^2 + \mathbf{X}_i^{-1})^\eta.$$

For $i = 1$ we have $q^{-1} \leq \mathbf{X}_1 = X/K \leq X^\eta$, because $X^{1-\eta} < K \leq qX$. For $i = 2$ we have $1 \leq \mathbf{X}_2 = K^{\eta/(1-\eta)} \leq (qX)^{2\eta}$. In both cases the last bound is $\ll (qX)^{4\eta}$.

For $i = 1$, Theorem 6.1 therefore gives

$$\langle I | \Delta_q F_1 | \alpha \rangle \ll \Delta^{O(1)} (qXN)^{5\eta} \left(\frac{X}{N^{1/2}} \right)^{1/2} Z_0^\theta N^{1/2} = \Delta^{O(1)} (qXN)^{5\eta} X^{1/2} N^{1/4} Z_0^\theta.$$

Since $X \geq N^{1/2}$ and $N^{1/2} \geq 2H^{1/2}$, here

$$Z_0 \leq 1 + \frac{2EX/N^{1/2} + 1}{q} \leq 1 + \frac{3EX}{qN^{1/2}} = 1 + \frac{3X}{4dN^{1/2}} \leq 1 + \frac{X}{dH^{1/2}}.$$

For $i = 2$ we obtain in the same way, using $X_2 \leq (qX)^2$ and $Z_0 \leq 1 + 3EX_2/(qN^{1/2}) \leq (X_2/X)(1 + X/(dH^{1/2}))$,

$$\langle I | \Delta_q F_2 | \alpha \rangle \ll \frac{X}{X_2} \Delta^{O(1)} (qXN)^{9\eta} \left(\frac{X_2}{N^{1/2}} \right)^{1/2} \left(\frac{X_2}{X} \right)^\theta \left(1 + \frac{X}{dH^{1/2}} \right)^\theta N^{1/2}.$$

This is the bound for $i = 1$ multiplied by $(qXN)^{4\eta} (X/X_2)^{1/2-\theta} \leq (qXN)^{4\eta}$, as $\theta < 1/2$. By (6.5) and Step 1, and since $N \leq 8H$, the proposition follows after renaming η . $\square$

Remark 6.6. For $X < K \leq qX$ the moduli exceed the length of the ℓ -sum. Throughout the range $K \leq qX$, Proposition 6.5 gives square-root cancellation in X , with a loss $H^{1/4}$ that does not depend on K . The condition $K \leq qX$ ensures $q\mathbf{X}_1 \geq 1$, so that the choice $Z_1 = q$ costs nothing in Lemma 6.3. Proposition 6.5 is a single-level analogue, with a residue class condition on the moduli, of the case $D = 1$ of [9, Theorem 1.4], where the bound is $X^{1/2}(1 + h^{1/4})(1 + X/(1 + h^{1/2}))^\theta$. When the estimate is averaged over levels $d \leq D$, as in [9, Theorem 1.4], the total bound is $D^{1/2}X^{1/2}(D^{1/2} + h^{1/4})(\dots)^\theta$. For $h \geq D^2$ this saves a factor $D^{1/2}$ for each level, through the divisor-switching argument discussed in Remark 8.1. That saving is not available here, because our polynomials change with d .

7. THE ASYMPTOTIC FORMULA FOR $\Sigma_d(x)$

Throughout this section d is a good prime, $m = m_d$, $H = H_d$, $N = EH$, $q = 4Ed$, $0 < \varepsilon < 1/100$, and

$$(7.1) \quad d \leq x^{1/4}, \quad x \geq x_0(a, b, \varepsilon).$$

Proposition 7.1. *Under (7.1),*

$$\Sigma_d(x) = \frac{32 c_F \mathfrak{L}_d}{2^j} \sqrt{\frac{x}{d}} + O(x^{1/4+\varepsilon} d^{1/2+\varepsilon} + x^{1/4+\theta/2+\varepsilon} d^{3/4-3\theta/2}).$$

7.1. Set-up. For real t put $T(t) = 2^{v-2}(Et^2 + H)/d$, so that $T(\ell) = T_d(\ell)$, and let $X > 0$ be defined by $T(X) = x$. Since $T(0) = m/(4d) \leq d/2$,

$$(7.2) \quad \frac{dx}{2^{v-1}E} \leq X^2 \leq \frac{dx}{2^{v-2}E}.$$

In particular $X \asymp (dx)^{1/2}$, and $X \geq N^{1/2}$ because $N \leq 2Ed^2$. For $y > 0$ and $t \in \mathbb{R}$ put

$$\Phi_y(t) = F(T(t)/x) W\left(\frac{2^{(v-2)/2}y}{\sqrt{T(t)}}\right).$$

Let $\ell \in \mathbb{Z}$ with $d \mid E\ell^2 + H$. By §4, $T(\ell) = 2^{v-2}T'$ with $T' \equiv 1 \pmod{4}$, so Lemma 5.1 applies with $s = v - 2$. For odd k we have $k \mid T(\ell)$ if and only if $k \mid T'$, that is, if and only if $dk \mid E\ell^2 + H$, since $E\ell^2 + H = dT'$. Hence

$$(7.3) \quad \Sigma_d(x) = 8 \sum_{k \text{ odd}} \chi(k) \sum_{\substack{\ell \in \mathbb{Z} \\ dk \mid E\ell^2 + H}} \Phi_k(\ell).$$

Only $k < k_+ := 2^{1-(v-2)/2}\sqrt{x}$ contribute, because $W(t) = 0$ for $t \geq 2$ and $T(t) \leq x$ on the support of $F(T(t)/x)$. We put

$$(7.4) \quad \mathcal{M}_d = 8 \sum_{k \text{ odd}} \chi(k) \frac{\varrho(dk)}{dk} \int_{\mathbb{R}} \Phi_k(t) dt, \quad \mathcal{E}_d = \Sigma_d(x) - \mathcal{M}_d.$$

By Lemma 5.2(a), $\varrho(dk) = \varrho_{E,H}(dk)$ for odd k .

7.2. The main term.

Lemma 7.2. $\mathcal{M}_d = \frac{32 c_F \mathfrak{L}_d}{2^j} \sqrt{\frac{x}{d}} + O(x^{1/4+\varepsilon} d^{1/2+\varepsilon})$.

Proof. Since $\varrho(dk) = 2f(k)$ for odd k and the sum over k is finite,

$$\mathcal{M}_d = \frac{16}{d} \int_{\mathbb{R}} F(T(t)/x) G(2^{-(v-2)/2} \sqrt{T(t)}) dt, \quad G(y) := \sum_{k \geq 1} \frac{\chi(k)f(k)}{k} W(k/y).$$

By (5.1), partial summation and Lemma 5.3(b),

$$\mathfrak{L}_d - G(y) = - \int_{y/2}^{\infty} A(u) \frac{d}{du} \left(\frac{1 - W(u/y)}{u} \right) du \ll y^{-1/2+\varepsilon} d^{1+\varepsilon} \quad (y \geq 1).$$

On $t > 0$ we substitute $T = T(t)$. Since $2^{v-1}E = 2^{2j-1}$ and $2^{v-2}H = m/4$, we have $dT/dt = 2^{2j-1}t/d$ and $2^j t = \sqrt{4dT - m}$, so $dt = 2^{1-j}d \, dT / \sqrt{4dT - m}$. As the integrand is even in t ,

$$\mathcal{M}_d = \frac{64}{2^j} \int_{x/2}^x \frac{F(T/x) G(\sqrt{T/2^{v-2}})}{\sqrt{4dT - m}} dT.$$

Replacing G by $\mathfrak{L}_d$ costs $\ll 2^{-j} \int_{x/2}^x x^{-1/4+\varepsilon} d^{1+\varepsilon} (dT)^{-1/2} dT \ll x^{1/4+\varepsilon} d^{1/2+\varepsilon}$. Replacing $(4dT - m)^{-1/2}$ by $(4dT)^{-1/2}$ costs $\ll |\mathfrak{L}_d| x m (dx)^{-3/2} \ll d^{3/2+\varepsilon} x^{-1/2}$, using $|\mathfrak{L}_d| \ll d^{1+\varepsilon}$. Finally $\int F(T/x) (4dT)^{-1/2} dT = c_F \sqrt{x} / (2\sqrt{d})$. $\square$

7.3. The error term.

Lemma 7.3. $\mathcal{E}_d \ll x^{1/4+\theta/2+\varepsilon} d^{3/4-3\theta/2}$.

Proof. Fix a smooth $\psi_0 \geq 0$ supported in $[1, 2]$ with $\sum_{i \in \mathbb{Z}} \psi_0(t/2^{i/2}) = 1$ for all $t > 0$. For instance, take a smooth $\phi \geq 0$ supported in $[1, 2]$ and positive on $(1, 2)$, and put $\psi_0 = \phi / \sum_i \phi(\cdot/2^{i/2})$. Let $\mathcal{K} = \{2^{i/2} : i \in \mathbb{Z}, 2^{i/2} \geq 1/2\}$, so that $\sum_{K \in \mathcal{K}} \psi_0(k/K) = 1$ for every integer $k \geq 1$. Writing $\chi(k) = \sum_{\kappa \in \{1,3\}} \chi(\kappa) \mathbf{1}\{k \equiv \kappa \pmod{4}\}$, we get from (7.3) and (7.4)

$$\mathcal{E}_d = 8 \sum_{K \in \mathcal{K}} \sum_{\kappa \in \{1,3\}} \chi(\kappa) \mathcal{E}(K, \kappa),$$

where

$$\mathcal{E}(K, \kappa) = \sum_{k \equiv \kappa \pmod{4}} \psi_0\left(\frac{k}{K}\right) \left(\sum_{\substack{\ell \in \mathbb{Z} \\ dk \mid E\ell^2 + H}} \Phi_k(\ell) - \frac{\varrho(dk)}{dk} \int_{\mathbb{R}} \Phi_k(t) dt \right).$$

If $K \geq k_+$ then $\mathcal{E}(K, \kappa) = 0$, because $\Phi_k = 0$ for $k \geq k_+$. So only $O(\log x)$ values of K contribute. Put $k_- = 2^{-3-(v-2)/2}\sqrt{x}$.

Write $\mu = dk$. Then $k \equiv \kappa \pmod{4}$ if and only if $\mu \equiv \kappa d \pmod{4d}$, and $\psi_0(k/K) = \psi_0(\mu/(dK))$. Let $\psi_2(s) = F(T(Xs)/x)$. Since $T(Xs)/x = \alpha s^2 + 1 - \alpha$ with $\alpha = EX^2/(EX^2 + H) \in [\frac{1}{2}, 1]$, the function ψ_2 is supported in $[-1, 1]$, and its derivatives are bounded in terms of F .

The case $K \leq k_-$. If $\psi_0(k/K) \neq 0$ and $F(T(t)/x) \neq 0$, then $2^{(v-2)/2}k/\sqrt{T(t)} \leq 2^{(v-2)/2} \cdot 2k_-/\sqrt{x/2} = 2^{-3/2} < 1/2$. Hence $W(\cdot) = 1$ and $\Phi_k(t) = \psi_2(t/X)$. Thus $\mathcal{E}(K, \kappa)$ is the sum Ξ of Proposition 6.5, with K replaced by dK , $\psi_1 = \psi_0$, ψ_2 as above and $\Delta = 1$.

The case $k_- < K < k_+$. Let $\widetilde{W}(w) = \int_0^\infty W(y)y^{w-1}dy$ for $\operatorname{Re} w > 0$. Since W' is supported in $[\frac{1}{2}, 2]$, integration by parts gives $\widetilde{W}(w) = -w^{-1} \int W'(y)y^w dy$, and repeating it gives $|\widetilde{W}(w)| \ll_A (1 + |w|)^{-A}$ for $\operatorname{Re} w = 1$. By Mellin inversion, $W(y) = \frac{1}{2\pi i} \int_{(1)} \widetilde{W}(w)y^{-w}dw$ for $y > 0$. Hence, whenever $\psi_0(k/K) \neq 0$,

$$\psi_0\left(\frac{k}{K}\right)\Phi_k(t) = \frac{1}{2\pi i} \int_{(1)} \widetilde{W}(w) \psi_{1,w}\left(\frac{k}{K}\right) \psi_{2,w}\left(\frac{t}{X}\right) dw,$$

where

$$\psi_{1,w}(s) = \psi_0(s)s^{-w}, \quad \psi_{2,w}(s) = \psi_2(s)(\alpha s^2 + 1 - \alpha)^{w/2} \left(\frac{\sqrt{x}}{2^{(v-2)/2}K}\right)^w.$$

For $\operatorname{Re} w = 1$ we have $|(\sqrt{x}/(2^{(v-2)/2}K))^w| = \sqrt{x}/(2^{(v-2)/2}K) \leq 8$, and $\alpha s^2 + 1 - \alpha \in [\frac{1}{2}, 1]$ on the support of ψ_2 . So $\|\psi_{i,w}^{(\nu)}\|_\infty \ll_\nu (1 + |w|)^\nu$ for $i = 1, 2$. We insert this into $\mathcal{E}(K, \kappa)$, interchange the finite sums with the integral, and use $\int_{\mathbb{R}} \psi_{2,w}(t/X) dt = X \int_{\mathbb{R}} \psi_{2,w}$. This gives

$$\mathcal{E}(K, \kappa) = \frac{1}{2\pi i} \int_{(1)} \widetilde{W}(w) \Xi_w dw,$$

where Ξ_w is the sum of Proposition 6.5 with K replaced by dK , $\psi_1 = \psi_{1,w}$, $\psi_2 = \psi_{2,w}$ and $\Delta = 1 + |w|$.

Application of Proposition 6.5. The hypotheses hold. By §4, H is odd, $d \nmid H$, and N is neither a square nor three times a square. We have $X \geq N^{1/2}$, and $1 \leq dK \leq dk_+ \leq qX$ by (7.2). Moreover $q \ll d$, $H \asymp d^2$, and by (7.2) and (7.1)

$$X^{1/2}H^{1/4} \asymp x^{1/4}d^{3/4}, \quad \frac{X}{dH^{1/2}} \asymp x^{1/2}d^{-3/2} \geq 1.$$

Proposition 6.5 with η small in terms of ε therefore gives, in both cases,

$$\Xi_w \ll (1 + |w|)^C x^{1/4+\theta/2+\varepsilon/2} d^{3/4-3\theta/2}$$

for some constant C , and the same bound for Ξ . Integrating against $|\widetilde{W}(w)|$ and summing over the $O(\log x)$ values of K and the two values of κ proves the lemma. $\square$

Proposition 7.1 follows from Lemmas 7.2 and 7.3.

Proof of Theorem 4.3. Let $\varpi < \varpi_\theta$ and let $d \leq x^\varpi$ be a good prime. Since $\varpi_\theta \leq 1/5$, (7.1) holds. By Lemma 5.3(d), the main term in Proposition 7.1 is $\geq c(\varepsilon) x^{1/2}d^{-1/2-\varepsilon}$. The error terms are smaller by a factor

$$\ll x^{-1/4+\varepsilon}d^{1+2\varepsilon} + x^{-1/4+\theta/2+\varepsilon}d^{5/4-3\theta/2+\varepsilon} \leq x^{-1/4+\varepsilon+\varpi(1+2\varepsilon)} + x^{-1/4+\theta/2+\varepsilon+\varpi(5/4-3\theta/2+\varepsilon)}.$$

Both exponents are negative once ε is small enough in terms of ϖ and θ : the first because $\varpi < 1/4$, and the second because $\varpi(5/4 - 3\theta/2) < 1/4 - \theta/2$. $\square$

This completes the proof of Theorem 1.1.

8. REMARKS

Remark 8.1 (The exponent and the main obstruction). The exponent comes from comparing the error term $x^{1/4+\theta/2}d^{3/4-3\theta/2}$ in Proposition 7.1 with the main term $x^{1/2}d^{-1/2}$. In Proposition 6.5 the factor $H^{1/4} \asymp d^{1/2}$ comes from the bound $N^{1+\eta}$ in Lemma 6.4. For the count of pairs in Step 3 of its proof this bound is essentially sharp. Since a disc $\{w : u(w, z) \leq 1\}$ has hyperbolic area 4π and a fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$ has area $\pi/3$, the equidistribution of Heegner points [3] suggests that there are about $12(\#\Lambda_N)^2 = N^{1+o(1)}$ pairs $(\mathfrak{z}, \mathfrak{w}) \in \Lambda_N \times \mathcal{Q}_N$ with $u(z(\mathfrak{w}), z(\mathfrak{z})) \leq 1$, and the computations in §9 confirm this.

The kernel $\langle \alpha | K_q k | \alpha \rangle$ itself is much smaller. In Step 2 of the proof of Lemma 6.4 we discarded the condition $\tau_1 g \diamond \mathfrak{z}_2 \in \mathcal{Q}$. Together with $\tau_1 \diamond \mathfrak{z}_1 \in \mathcal{Q}$, it implies that the binary forms of $\mathfrak{z}_1$ and $\mathfrak{w} = g \diamond \mathfrak{z}_2$ have a common zero in $\mathbb{P}^1(\mathbb{F}_d)$, and this happens for only about $4/d$ of the pairs (§9). In the examples of §9, the exact value of $\langle \alpha | K_q k | \alpha \rangle$ for $k = \mathbf{1}\{u_1 \leq 1\}$ is at most twice the diagonal contribution $\langle 1 \rangle_\alpha$, which is $O(N^{1/2+o(1)})$. In [9], where the polynomial is fixed and the level varies, the analogous congruence is exploited by summing over the level with a divisor bound [9, Proposition 4.1]; at a single level this is not possible. A bound $N^{1+\eta}/d + N^{1/2+\eta}$ in Lemma 6.4 would improve Proposition 6.5 by a factor $d^{-1/2}$. Together with an easy improvement of the error term in Lemma 7.2, this would give $\varpi < 1/3$ and $\delta < 1/6$ for every $\theta < 1/2$. For that improvement, shift the contour in the Mellin integral representing $G(y)$ to $\mathrm{Re} w = -1/2 + \varepsilon$ and use the convexity bound for $L(s, \psi)$; this gives $\mathcal{L}_d - G(y) \ll y^{-1/2+\varepsilon} d^{1/2+\varepsilon}$. A bound of this kind for the kernel amounts to a pair-correlation statement for Heegner points of discriminant $-4N$ on $X_0(d)$ at the scale of their mean spacing. This seems out of reach; compare [3] for equidistribution at a fixed scale.

The exponent $1/6$ is in any case the limit for arguments that treat the primes d separately. For $d > x^{1/3}$ the polynomial T_d takes fewer than d values up to x , fewer than the number $h(-4N_d) = N_d^{1/2+o(1)}$ of classes of forms that govern its roots. Going beyond requires averaging over d . For the patterns with $B = 1$, such as $\{0, 1, 2\}$, this is a question about the equidistribution, in residue classes, of norms of hyperbolic lattice points; see [11, Remarks 8.1 and 8.5] and [7].

Remark 8.2 (Why the hyperbola method is kept). One might try to evaluate $\Sigma_d(x)$ without Lemma 5.1, by applying Proposition 6.5 to all divisors of $T' = (E\ell^2 + H)/d$. The moduli would then go up to about X^2 , far beyond the range $K \leq qX$ of Proposition 6.5, and in that range no estimate of this kind can hold. Indeed, the moduli $\mu = E\ell^2 + H$, with complementary divisor 1, all satisfy $\mu \equiv H \pmod{4}$, so they lie in only one of the two classes κ . For $K \asymp X^2$ about X/d of them lie in $[K, 2K]$ and are divisible by d , which is the size of the expected main term. The hyperbola identity avoids such moduli. It needs T to be 2^{v-2} times a number $\equiv 1 \pmod{4}$ for all ℓ in a lattice, and this is what forces the 2-adic analysis of §3.

Remark 8.3 (Weyl sums uniformly in the discriminant). Duke, Friedlander and Iwaniec [5, Theorem 1.1] bound the sums

$$\sum_{c \equiv 0 \pmod{q}} f(c) \sum_{\substack{b \bmod c \\ b^2 \equiv D \pmod{c}}} e\left(\frac{hb}{c}\right),$$

for f supported on $[Y, 2Y]$, by $h^{1/4}(Y + h\sqrt{D})^{3/4} D^{1/8-1/1331}$. The bound is uniform in the positive fundamental discriminant D and in q , and by [5, §16] the same holds for negative fundamental discriminants. Its strength is the uniformity in D , which makes it non-trivial even for Y somewhat smaller than $\sqrt{D}$. In our application the moduli are odd, and the relevant discriminant is $-EH_d$, that is, $-4H_d$ or $-8H_d$. It has size d^2 , and it is fundamental exactly when H_d is squarefree, although Duke, Friedlander and Iwaniec note that non-fundamental discriminants can also be treated. The moduli go up to $d\sqrt{x}$, far beyond $\sqrt{D}$. In this range the bound saves the same

power $Y^{1/4}$ as the estimate obtained from Weil's bound in [11, Proposition 6.3], so it would not improve the exponent of [11] significantly. The method of [4, 5], in which the roots are parametrised by quadratic forms and the resulting automorphic kernels are analysed spectrally, is the starting point of [8, 9] and hence of the present paper.

Remark 8.4 (Effectivity). Siegel's theorem enters only through Lemma 5.3(d), and it can be avoided. Let s be the squarefree part of m_d . By the Landau–Page theorem, at most one primitive real character ψ' of conductor $\ll x^{2\varpi}$ has $L(1, \psi') < c_{\text{LP}}/\log x$, for a suitable absolute constant $c_{\text{LP}} > 0$. The primitive character attached to d depends only on s . For fixed s , the equation $(d+c)^2 - sy^2 = -4A(B-A)$ has $O(\log x)$ solutions with $0 < d \leq x$, by the theory of Pell equations. Discarding these $O(\log x)$ primes d leaves $\mathfrak{L}_d \gg (\log x)^{-1}(\log \log x)^{-2}$ effectively. The other inputs, in particular Theorem 6.1 and the Kim–Sarnak bound, are effective.

Remark 8.5 (Correlations of r). Let n arise from some (d, ℓ) in (4.3). By the proof of Lemma 2.5, no prime $\ell' \equiv 3 \pmod{4}$ divides both P and $P+B$. Since $r/4$ is multiplicative, it follows that $r(P(P+B)) \leq \frac{1}{4}r(P)r(P+B)$. Combined with Lemma 2.6 and Proposition 4.2, this gives

$$\sum_{d \text{ good}} \Sigma_d(x) \leq \frac{1}{4} \sum_{n \leq x} r(n) r(n+a) r(n+b).$$

Hence $\sum_{n \leq x} r(n)r(n+a)r(n+b) \gg x^{1/2+\delta}$ for every $\delta < \delta_\theta$. Conjecturally this correlation is $\asymp x$.

9. NUMERICAL CHECKS

The computations reported here were carried out with the scripts that accompany this manuscript.

The 2-adic set-up. For each of the 3725 patterns $\{0, a, b\}$ with $b \leq 100$ and a or b odd, we chose (p_0, B, A) by Table 1, constructed n_* as in the proof of Lemma 3.1, and computed t_* , d_* , v and d_0 as in the proof of Proposition 3.2. We then checked the conclusion of Proposition 3.2 for 40 random pairs (d, u) with $d \equiv d_0 \pmod{2^J}$ and $u \equiv 0 \pmod{2^j}$ for each pattern. There were no failures. Table 2 lists some of the data.

pattern	p_0	B	A	n_*	v	J	d_0	E	m_d
$\{0, 1, 2\}$	0	1	2	16	3	5	25	8	$d^2 + 6d + 1$
$\{0, 1, 3\}$	0	3	1	1	3	5	1	8	$d^2 - 2d + 9$
$\{0, 2, 3\}$	2	1	-2	20	3	5	21	8	$d^2 - 10d + 1$
$\{0, 1, 4\}$	0	1	4	16	4	6	25	4	$d^2 + 14d + 1$
$\{0, 3, 8\}$	0	3	8	77	2	4	5	4	$d^2 + 26d + 9$
$\{0, 5, 8\}$	0	5	8	320	5	7	53	8	$d^2 + 22d + 25$

TABLE 2. The data of Lemmas 2.3 and 3.1 and Proposition 3.2 for some patterns.

The main term. Table 3 compares the sums $\Sigma_d^+(x)$, defined as in (4.3) but over $\ell \geq 1$ only, with half of the main term of Proposition 7.1, at $x = 10^{13}$. For each pattern we took the first good prime $d > d_1$ and the first good prime $d > 1000$. We used $F(t) = \exp(-1/(20(t-\frac{1}{2})(1-t)))$ for $\frac{1}{2} < t < 1$, for which $c_F = 0.16684$. The constant $\mathfrak{L}_d$ was evaluated as $L(1, \chi)L(1, \psi)\mathcal{P}_d$ (Lemma 5.3(c)), with $L(1, \psi)$ computed from the rapidly convergent series supplied by the functional equation. The last column gives the corresponding ratio for the sharp cut-off, that is, for $\sum_{\ell \geq 1} r(T_d(\ell))$ over $T_d(\ell) \leq x$, compared with the same formula with c_F replaced by 2; this case is not covered by Proposition 7.1. In all these computations every $T_d(\ell)$ was an integer of the form $2^{v-2}T'$ with $T' \equiv 1 \pmod{4}$, as Proposition 3.2 predicts, and $P, P+B \in \mathcal{S}$ in every case, as Lemma 2.5 predicts.

pattern	d	$\#\{\ell\}$	$\Sigma_d^+(x)$	main term/2	ratio	sharp ratio
$\{0, 1, 2\}$	89	167 600	51 778	51 150	1.012	1.004
	1049	48 818	13 851	13 544	1.023	1.013
$\{0, 1, 3\}$	97	160 540	47 194	47 438	0.995	0.997
	1153	46 564	8 362	8 280	1.010	1.012
$\{0, 2, 3\}$	53	217 186	90 690	90 209	1.005	1.002
	1013	49 678	28 238	28 206	1.001	1.004
$\{0, 1, 4\}$	89	167 600	58 831	58 606	1.004	1.001
	1049	48 818	16 275	16 162	1.007	1.008
$\{0, 5, 8\}$	181	58 763	22 534	22 796	0.989	1.002
	1973	17 798	12 985	13 142	0.988	0.992

TABLE 3. The sums $\Sigma_d^+(x)$ at $x = 10^{13}$ compared with half of the main term $32c_F \mathfrak{L}_d 2^{-j} \sqrt{x/d}$ of Proposition 7.1. Here $\#\{\ell\}$ is the number of $\ell \geq 1$ with $d \mid E\ell^2 + H_d$ and $T_d(\ell) \leq x$.

Type I sums. We computed the sums Ξ of Proposition 6.5 for $(E, H) = (8, 1057)$ and $(8, 10081)$, which come from the pattern $\{0, 1, 2\}$ with $d = 89$ and $d = 281$, and for $(E, H) = (8, 285)$ and $(8, 3869)$, which come from $\{0, 2, 3\}$ with $d = 53$ and $d = 181$. We took $\psi_1(t) = \phi(2t - 3)$ and $\psi_2 = \phi$ with $\phi(t) = \exp(-1/(1 - t^2))$, $X = 2 \cdot 10^6$, $\kappa \in \{1, 3\}$, and $K = 2^{i/2}X$ for $0 \leq i \leq 2 \log_2 d$. In all cases $|\Xi| \leq 0.03 X^{1/2} H^{1/4}$; the maxima of $|\Xi|/(X^{1/2} H^{1/4})$ were 0.024, 0.003, 0.028 and 0.003. The largest values of $|\Xi|$ occur for K close to X . For $K \geq 4X$ we always found $|\Xi|$ at most the square root of the main term $\mathfrak{M}$, which is roughly the number of terms. This suggests that the loss $H^{1/4}$ in Proposition 6.5 is a feature of the method (Remark 8.1).

Pairs of Heegner points. For $N = EH_d$ with the pattern $\{0, 1, 2\}$ and $d \in \{89, 281, 409\}$, and with the pattern $\{0, 2, 3\}$ and $d \in \{53, 181, 373\}$, we counted the pairs $(\mathfrak{z}, \mathfrak{w}) \in \Lambda_N \times \mathcal{Q}_N$ with $u(z(\mathfrak{w}), z(\mathfrak{z})) \leq 1$; this is the quantity bounded in Step 3 of the proof of Lemma 6.4. The number of pairs lay between $4.5N$ and $61N$, and in each case it agreed with $12(\#\Lambda_N)^2$ to within 1.5%, as equidistribution predicts (Remark 8.1). The number of pairs whose binary forms have a common zero in $\mathbb{P}^1(\mathbb{F}_d)$ was between $3.6/d$ and $4.1/d$ times the number of all pairs. Finally, for $(E, H, d) = (8, 1057, 89)$, $(8, 285, 53)$ and $(8, 3869, 181)$ and both values of κ , we computed the exact value of $\langle \alpha | K_q k | \alpha \rangle$ in Lemma 6.4, for $\mathcal{Q} = \mathcal{Q}^\kappa$ and $k = \mathbf{1}\{u_1 \leq 1\}$, by means of the unfolding in Step 2 of its proof. The values are 416, 64 and 144, against the diagonal contributions $\langle 1 \rangle_\alpha = 224, 64$ and 144 and against $N = 8456, 2280$ and 30952. At most 8 cosets τ satisfied $\tau \diamond \mathfrak{z} \in \mathcal{Q}^\kappa$, against the bound $12E = 96$ of (6.3).

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